Paz Lab
Neural Mechanisms of Learning

Behavioral Neuroscience: Fear thou not

Rony Paz

What is the NeuroPsychological approach?

- **Neuropsychology** is the basic scientific discipline that studies the structure and function of the brain related to specific psychological processes and overt behaviors.
- The use of *artificial* well-controlled tasks.
- Why?
  - Well-controlled
  - Pre-planned quantification
  - We observed something, now lets do it properly
  - Target Human behavior
  - Statistically rigorous (being hypothesis-driven)
- Examples:
  - Sensation and perception
  - Control of movement
  - Learning and memory
  - Decision making
  - Emotions
  - Attention

Active and Passive approaches

- The active approach manipulates the brain and observes behavior.
  - Stimulation (electrical, optogenetics)
  - Lesions
  - Pharmacological intervention
  - Genetic manipulations

- The passive approach manipulates behavior and observes the brain.
  - Trained behaviors while measuring brain activity (electrophysiology, imaging 1P,2P,MRI)
  - Observing patients
  - Psychophysics / Behavioral experiments

- Correlation and Causation ?

Reward vs. Punishment

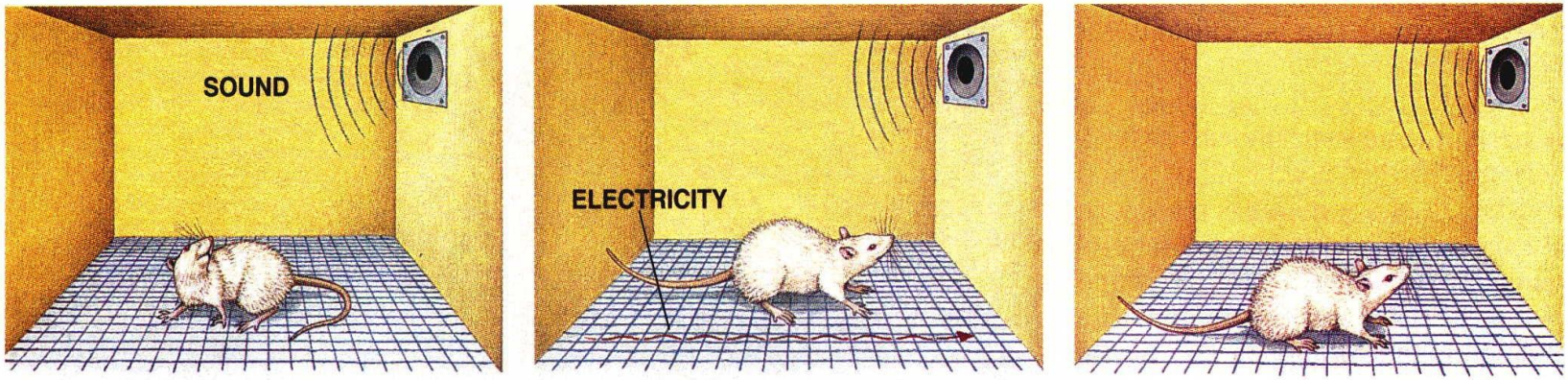
- What is a “reward”?
- Learning is best motivated by threats to survival?
- Threats are much better reinforcers?
- Fear is a prime motivator

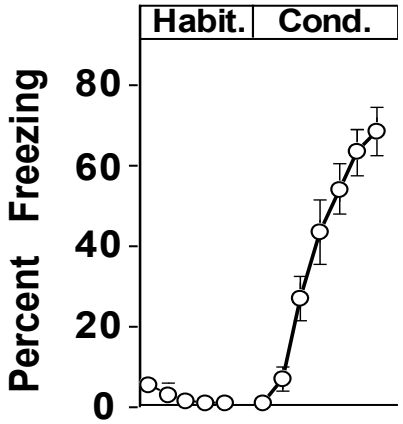
	Decreases behavior	Increases behavior
Presented	Positive punishment	Positive reinforcer
Taken away	Negative punishment	Negative reinforcer

Taking drugs?

More fun,
less withdrawal

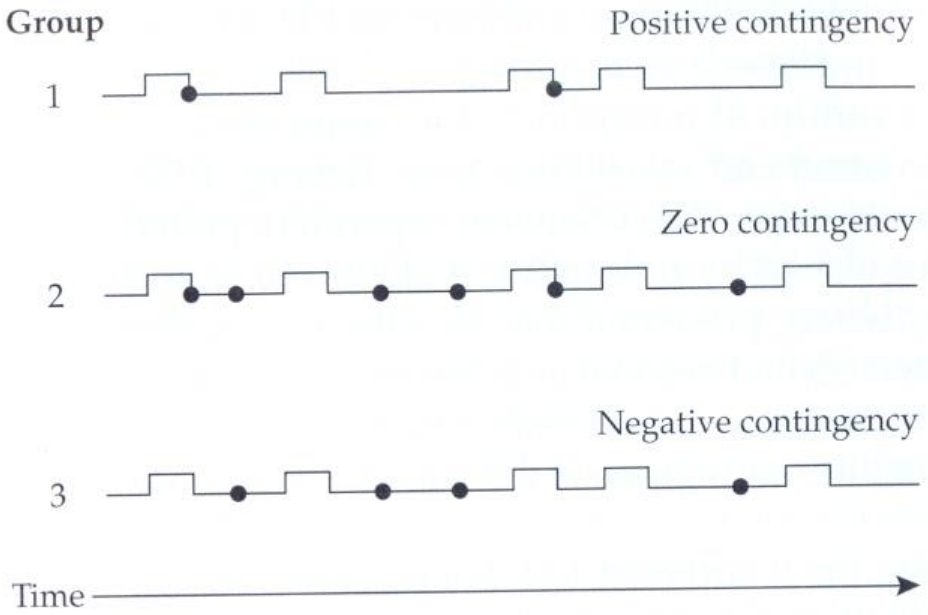
Classical fear conditioning

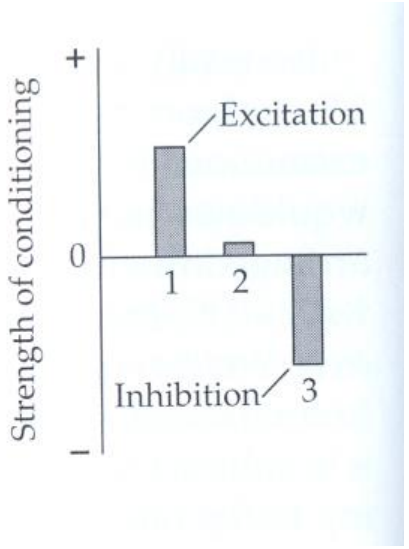




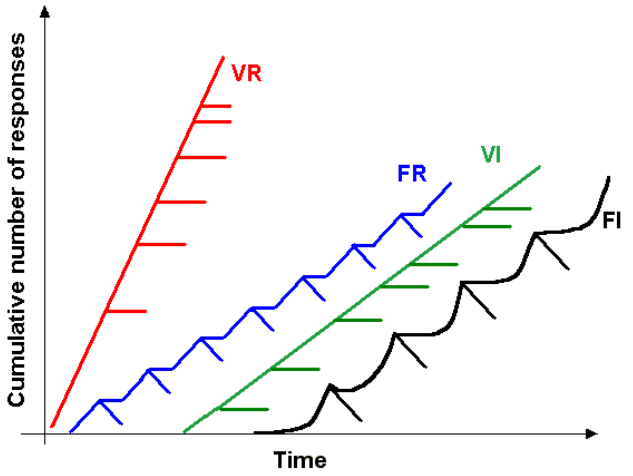
CS-US pairing
Tone = conditioned stimulus (CS)
Foot-shock = unconditioned stimulus (US)
Freezing = conditioned response (CR-UR)

Contingency: co-occurence

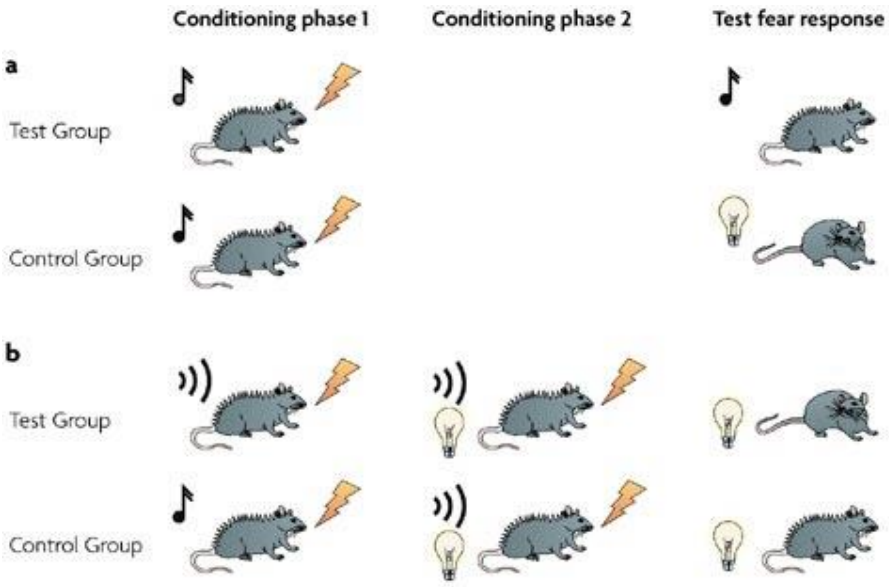




Schedules of reinforcement:
Variable/fixed interval/ratio



More than contingency: Surprise / added information



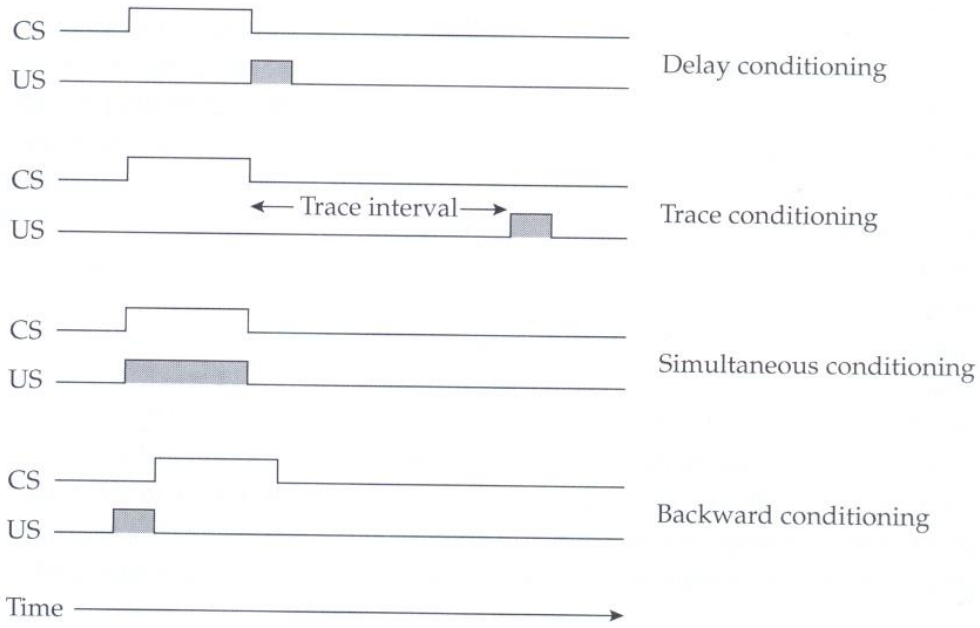
Aversive conditioning

Blocking



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Rules of thumb for conditioning strength

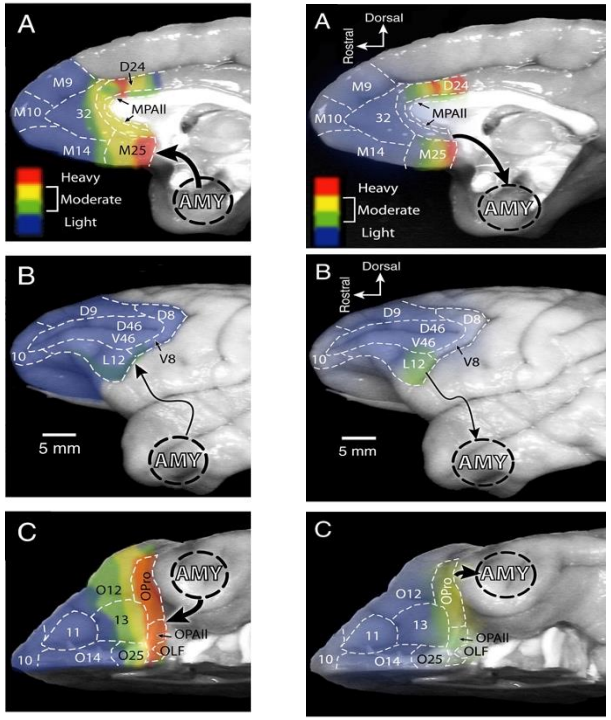


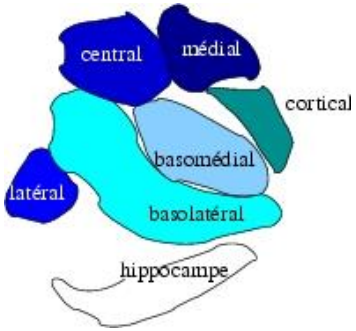
- Backward < simultaneous < trace < delay
- In trace: short interval > long interval
- In delay: short CS > long CS
- Salience of the CS
- Strength of the US
- Spaced trials is better than massed trials (the ratio between inter-trial-interval and the CS)

But notice it is hard to estimate backwards learning

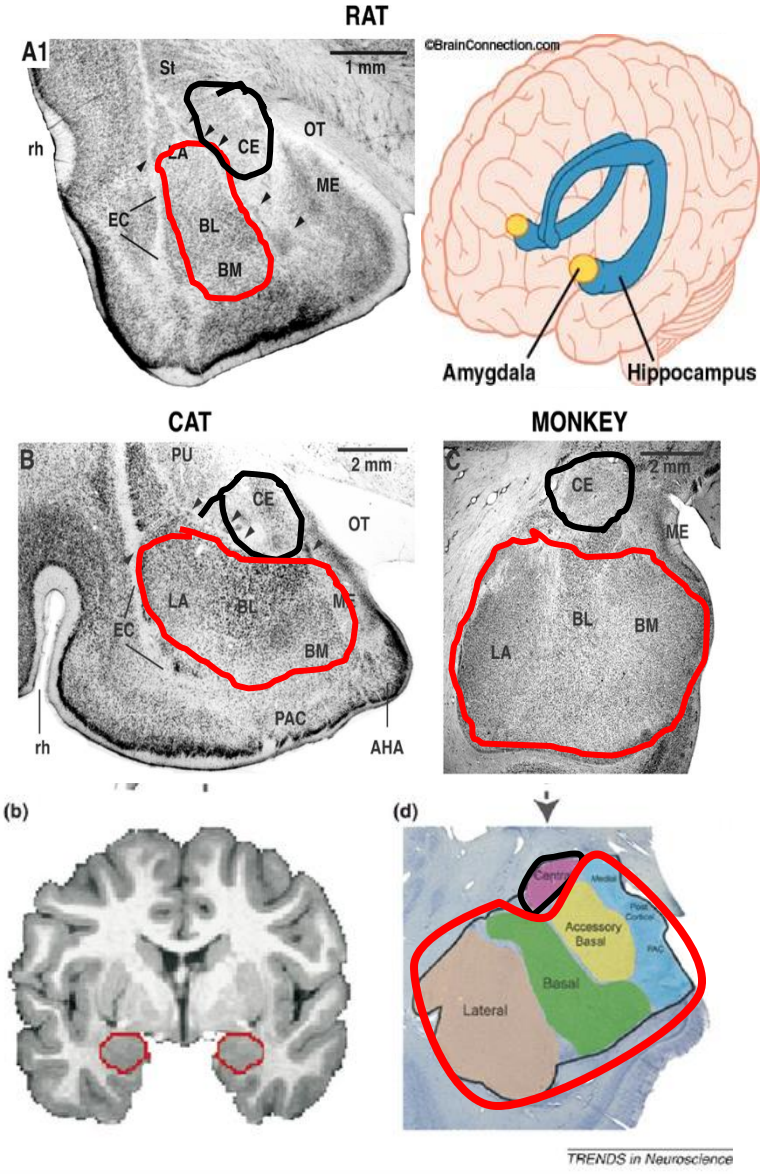
Amygdala

- Evolved in primates
- Correlated with the prefrontal cortex (PFC)
- Amygdala-PFC is a tight 2-way 1-synapse network
- Emotional responses and emotional learning
- Show abnormal activity in most psychiatric conditions, mainly in anxiety and mood disorders



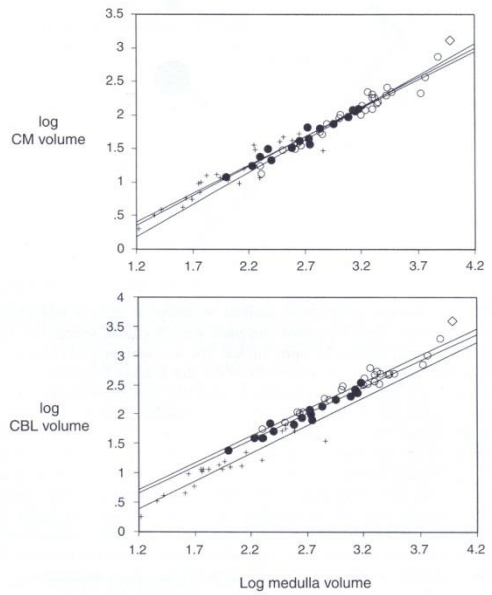


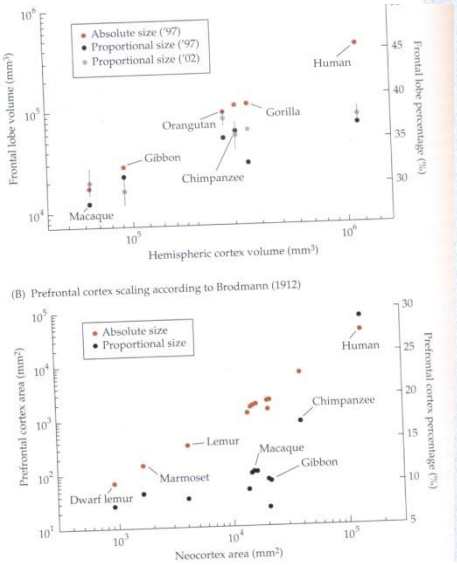
Ghashghaei et al. 2007

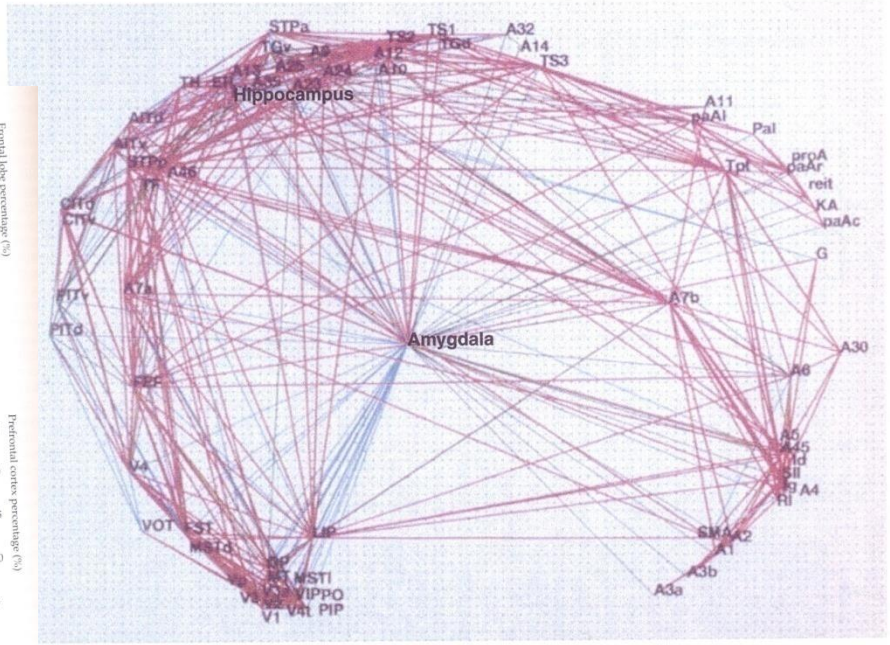


Amygdala and its basolateral complex (BLA)

- BLA evolution parallels that of the prefrontal cortex
- BLA cell types are similar to the cortex
- Cortical projections are much more extensive in primates
- Most cortical projections of the amygdala originate from BLA (none from CEA)





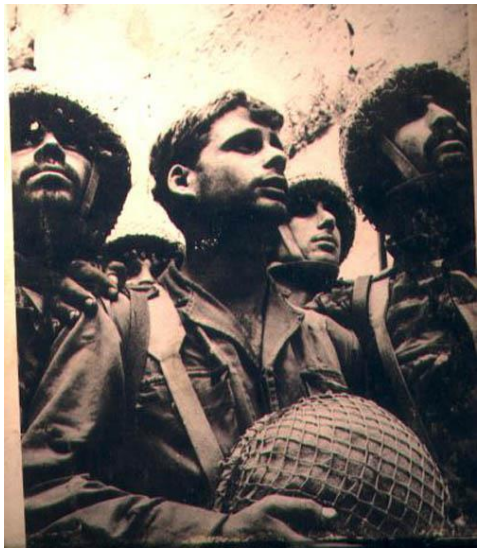


Humans and emotions



EMOTIONS FACILITATE MEMORY

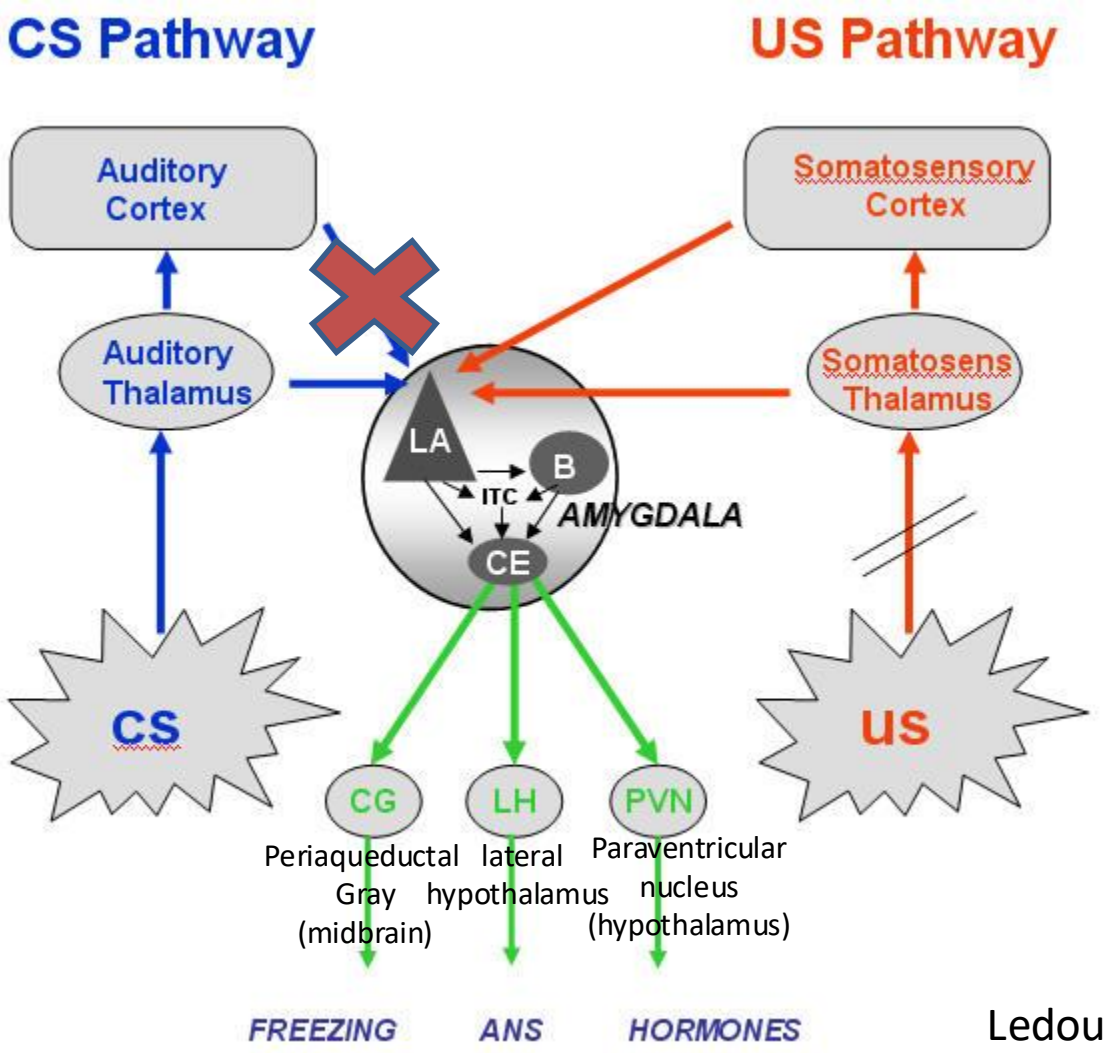
Emotionally arousing events are remembered more vividly.





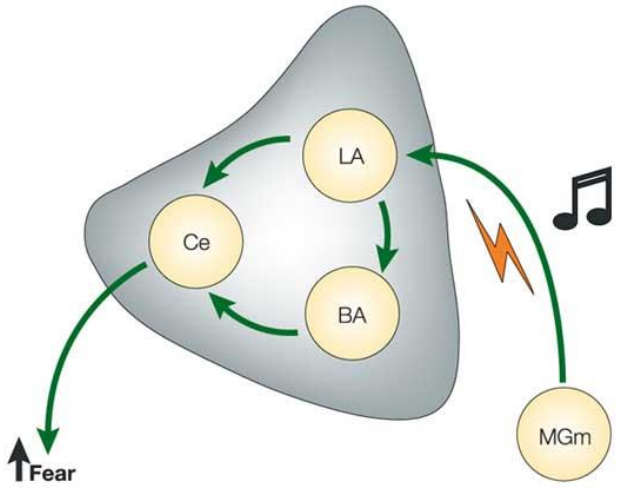


Fear circuit

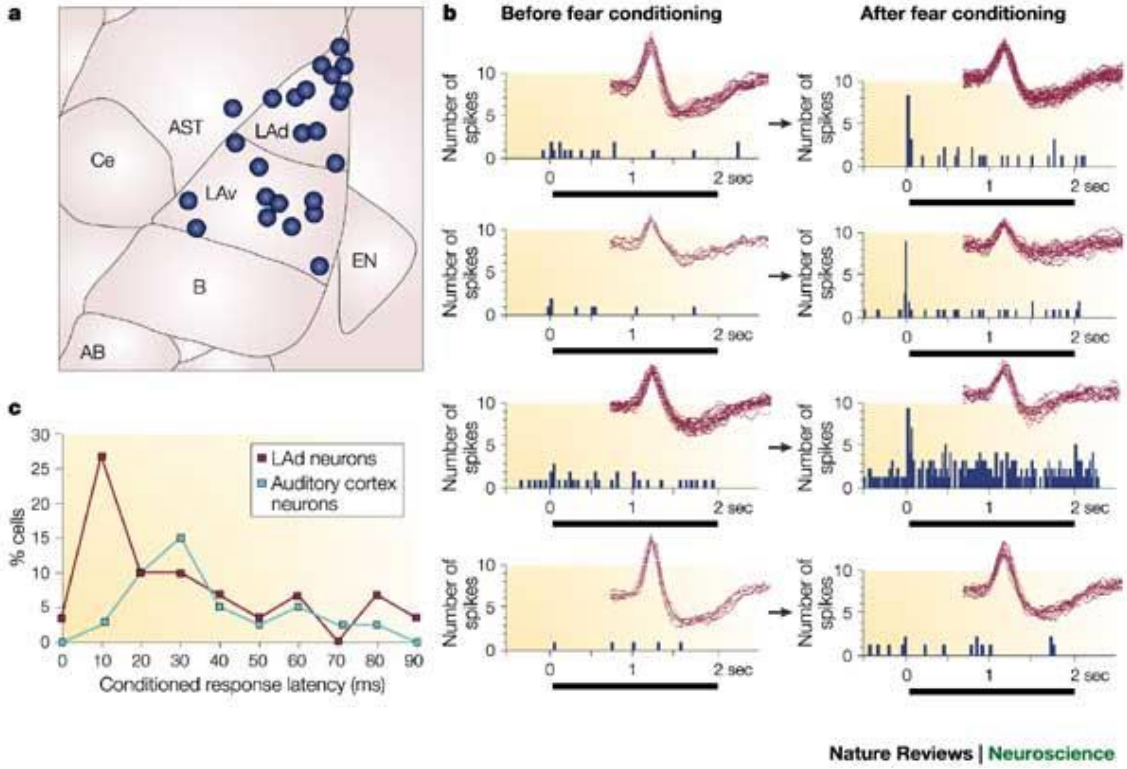


Ledoux, Mcgaugh, Davis

Neurons acquire tone responses after conditioning



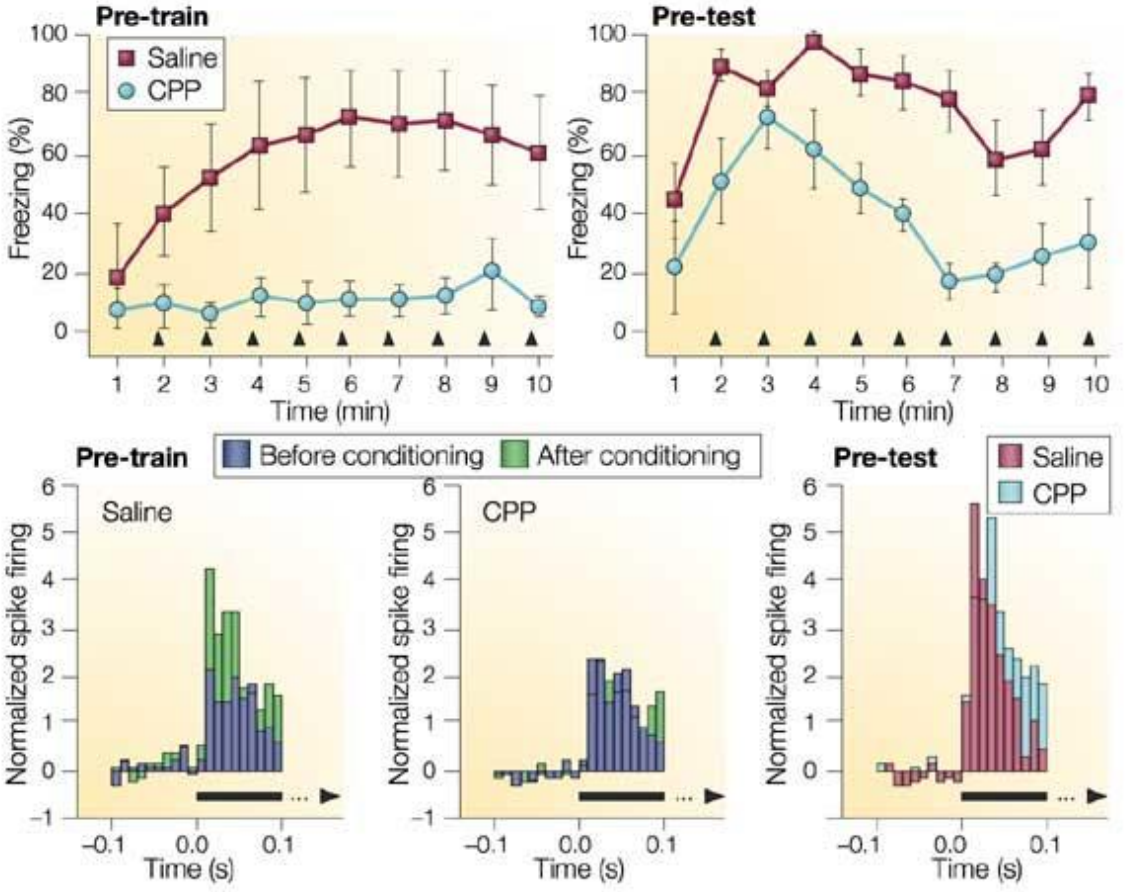
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LTP in the LA is required

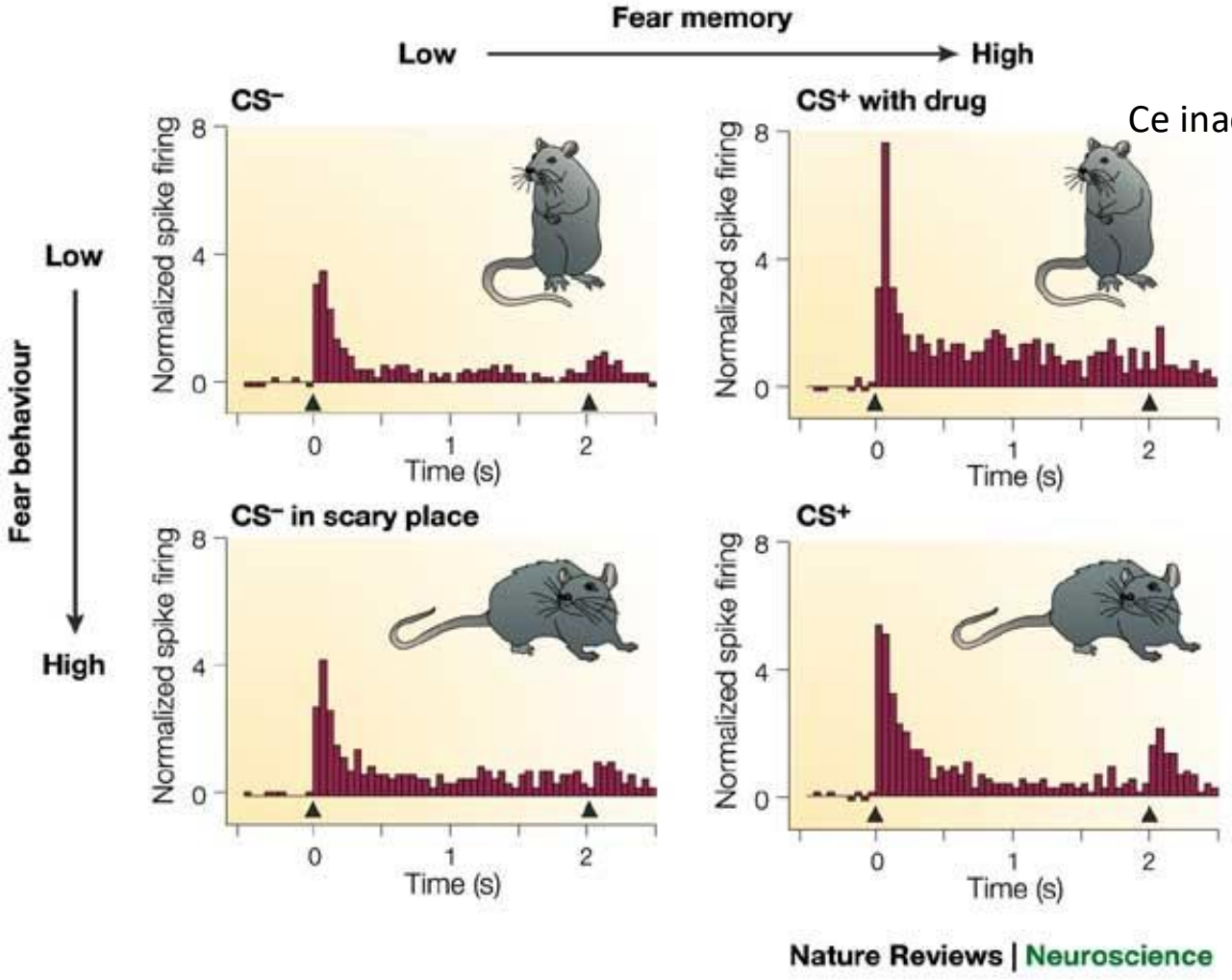
NMDA (***N***-methyl-**D**-aspartate, glutamate receptor) is involved in both the acquisition of fear memory and the induction of long-term potentiation (LTP) in the amygdala.



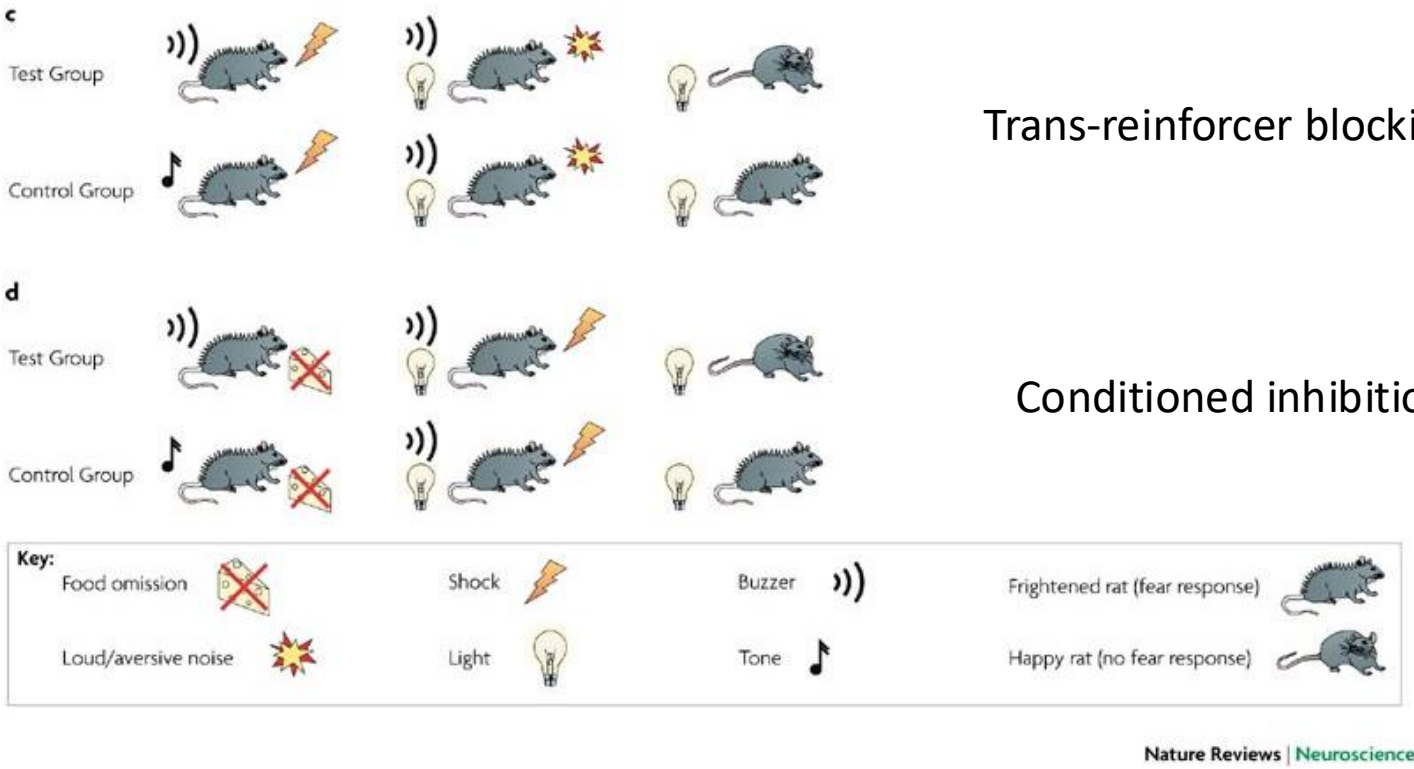
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CPP (3-(2-carboxypiperazin-4-yl) propyl-1-phosphonic acid), a competitive NMDA-receptor antagonist

LA encodes memory independent of fear behavior



Ce inactivation

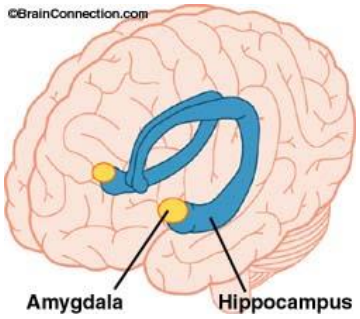


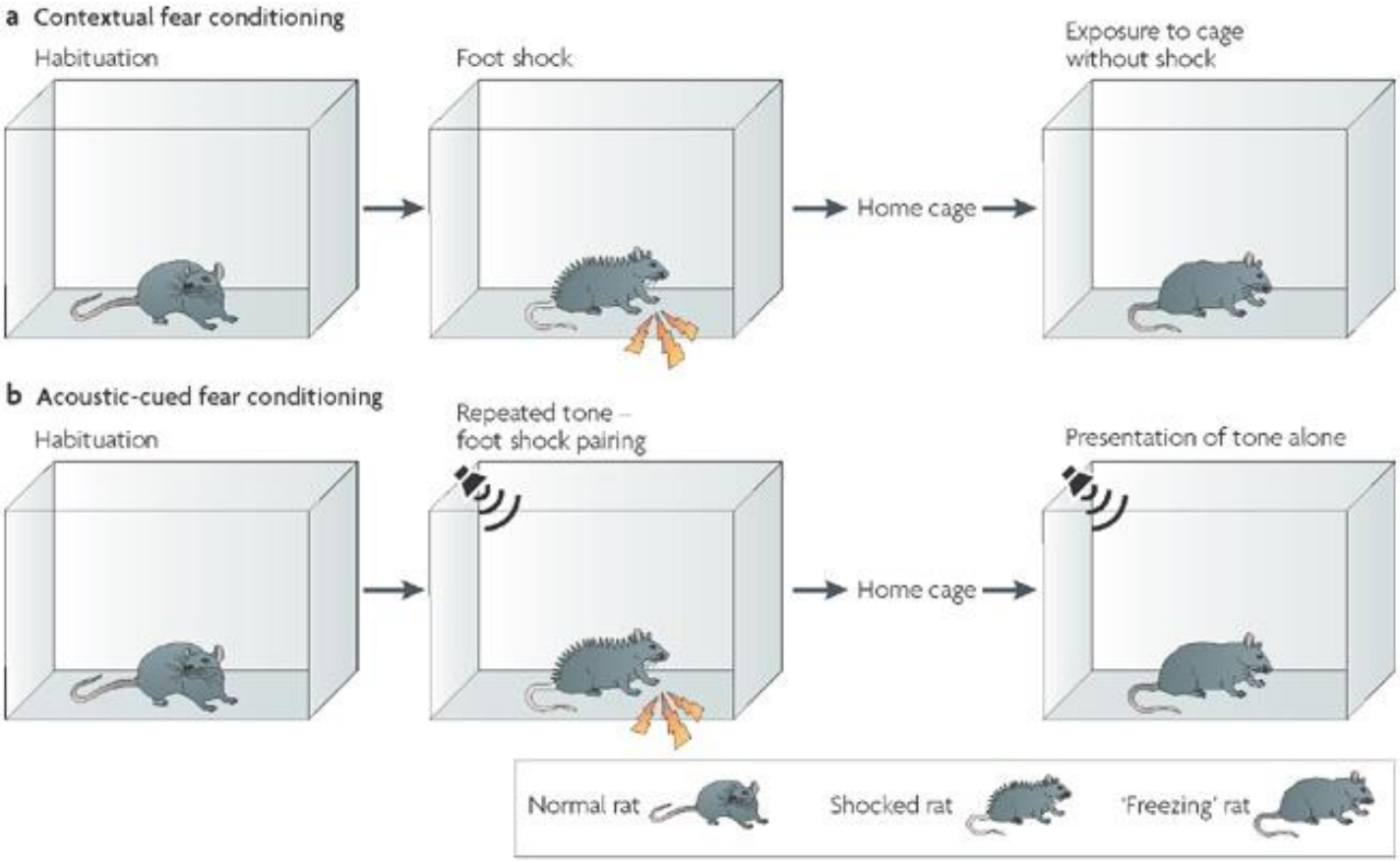
Trans-reinforcer blocking

Conditioned inhibition

Suggests common brain mechanisms

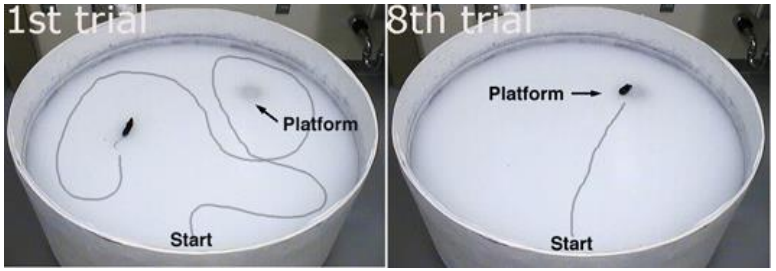
Contextual fear

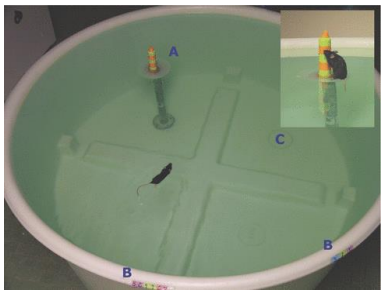




Amygdala modulation of memory

- Hippocampal dependent learning: spatial
- Striatum dependent-learning: cue-related





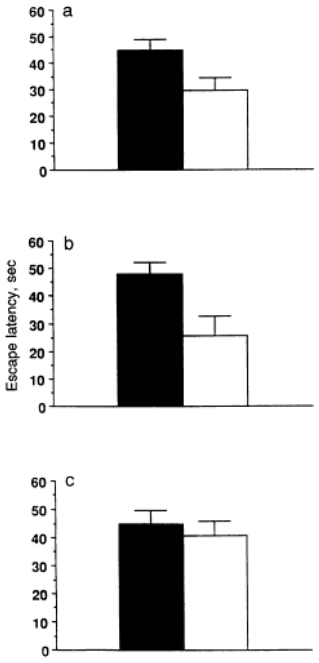


FIG. 1. Mean (±SE) escape latencies of *d*-amphetamine (10 μg) (□) and saline-treated (■) rats on the retention test trial in the spatial task. (*a*) Hippocampal injections. (*b*) Amygdala injections. (*c*) Caudate nucleus injections.

posttraining intracaudate and intrahippocampal injections of *d*-amphetamine on retention of cued and spatial learning in

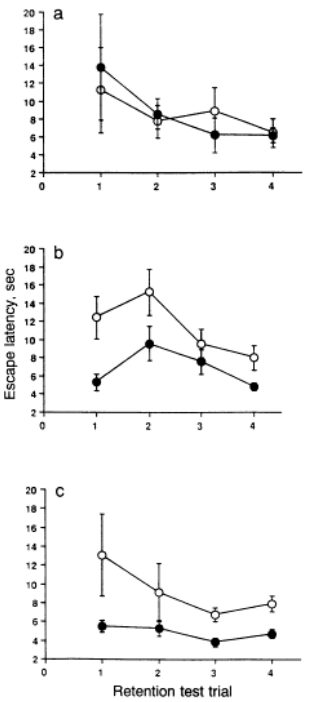


FIG. 2. Mean (±SE) escape latencies of *d*-amphetamine (10 μg) (●) and saline-treated (○) rats on the retention test trial in the cued task. (*a*) Hippocampal injections. (*b*) Amygdala injections. (*c*) Caudate nucleus injections.

Injection of d-amphetamine into the Amygdala affects both if right after training, but not if pre-testing

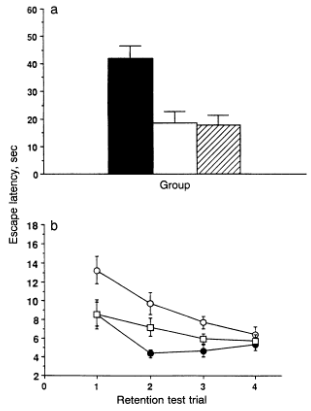


FIG. 3. Mean (±SE) escape latencies of rats receiving intraamygdala posttraining *d*-amphetamine or saline and rats receiving preretention test lidocaine or saline on the retention test trial(s) in the spatial task (*a*) and cued task (*b*). Posttraining/preretention: ■ (*a*) and ○ (*b*), saline/saline; □ (*a*) and ● (*b*), *d*-amphetamine/saline; ▨ (*a*) and □ (*b*), *d*-amphetamine/lidocaine.

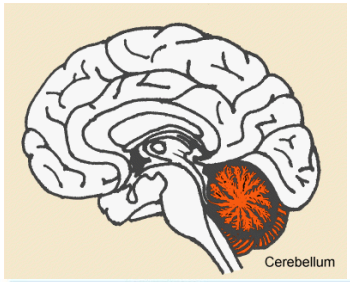
Packard, Mcgaugh

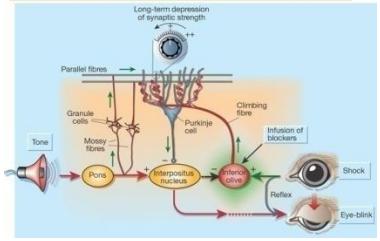
So, does it encode the memory or just modulates it?

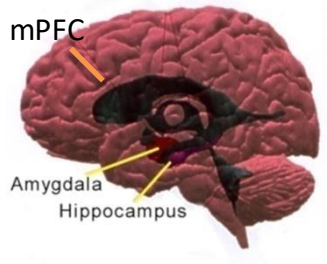
- **It depends.**

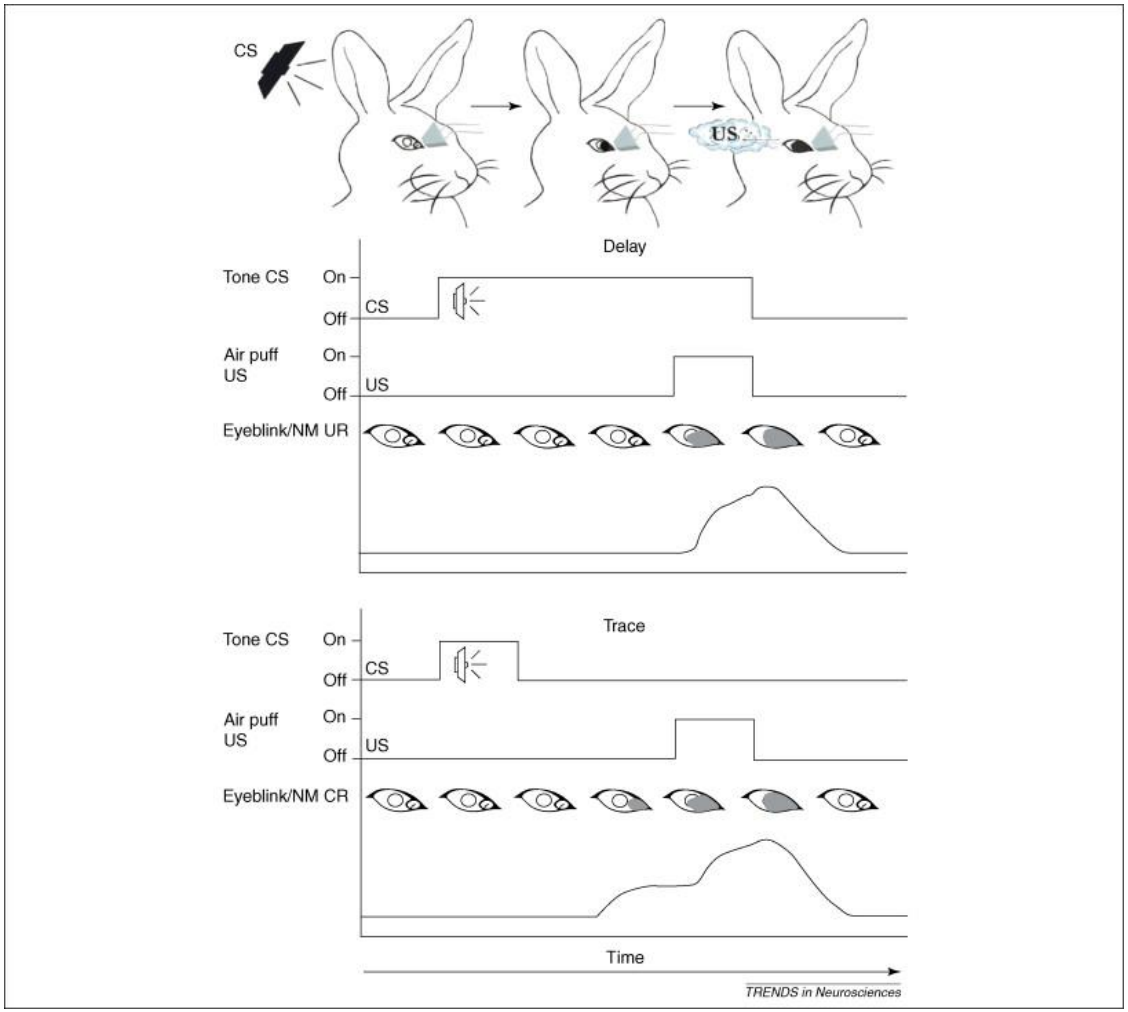
Eyelid (blink) reflex conditioning





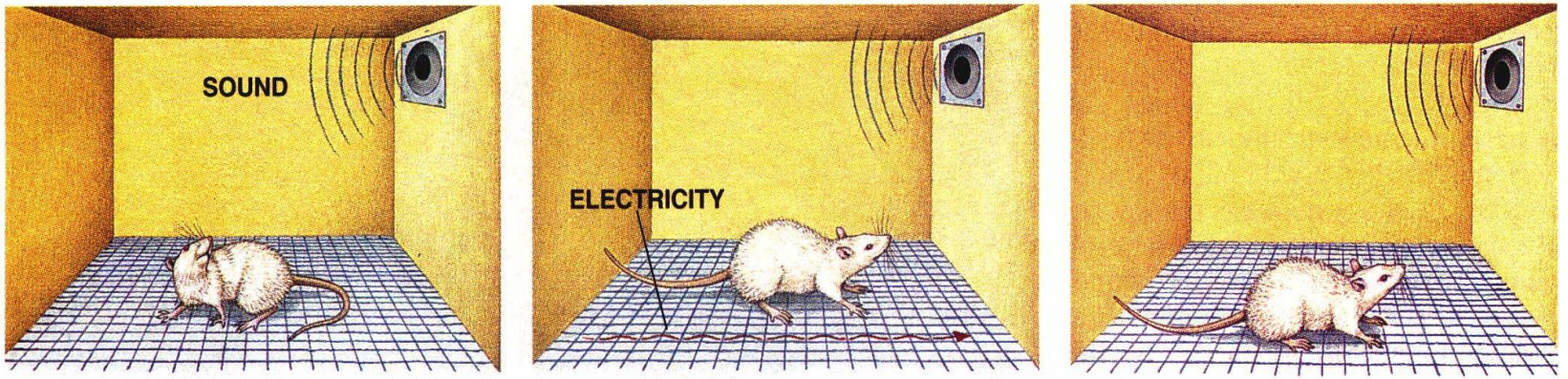


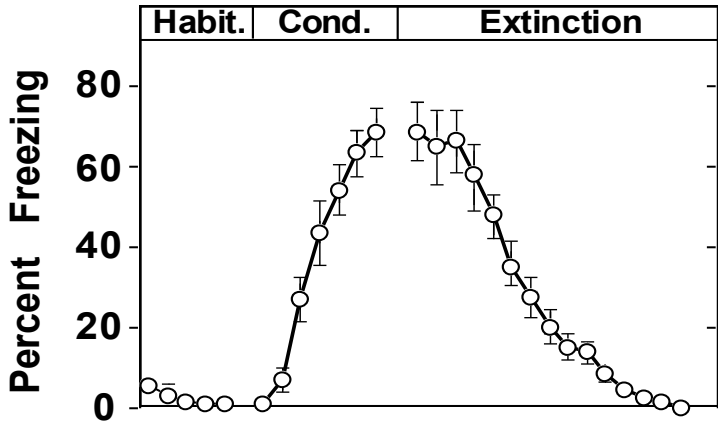




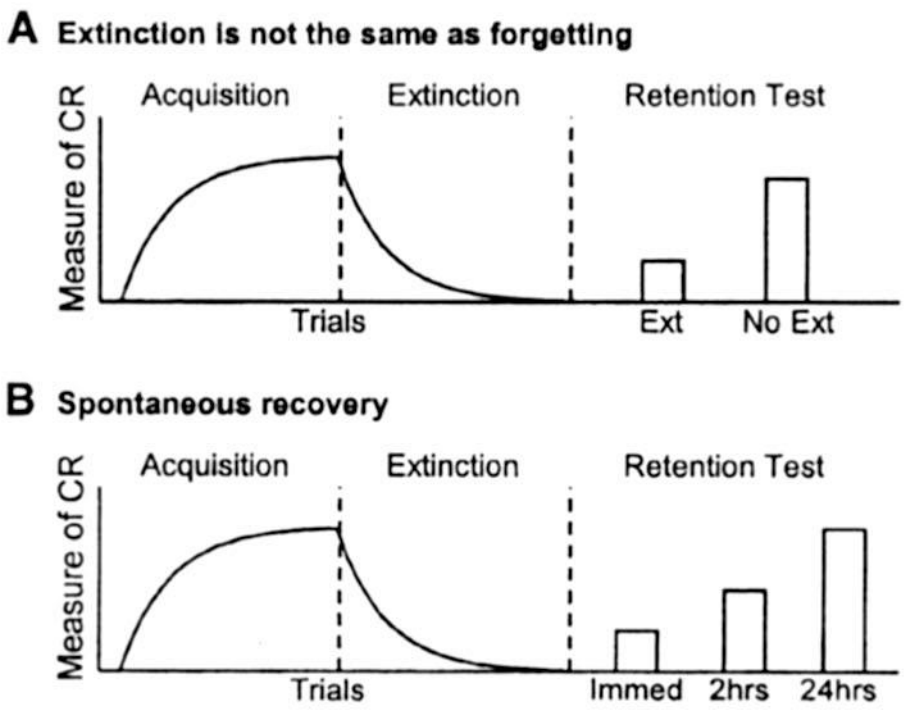
- Why is trace hippocampal-dependent?
- Maintaining the CS? Timing the trace? Harder?
- Eyelid requires ~0.3sec, and hippocampus is required when 0.5-1sec.
- In tone-shock, trace can be 3sec, and hippocampus is required for ~20sec
- This suggest context-conditioning

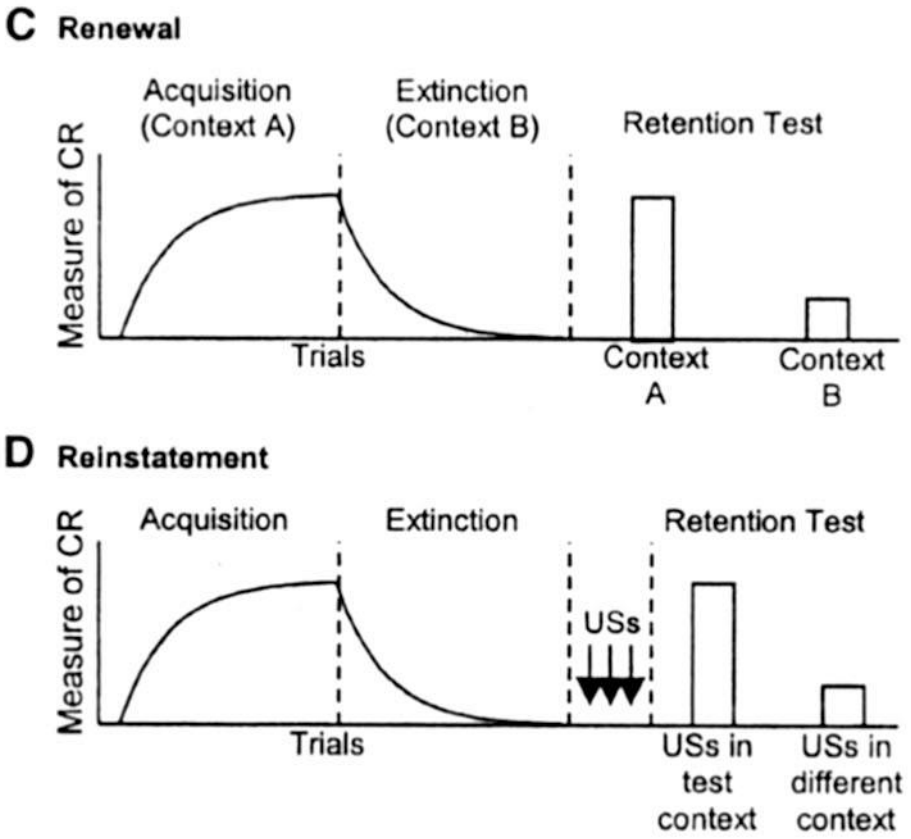
Extinction of fear-conditioning





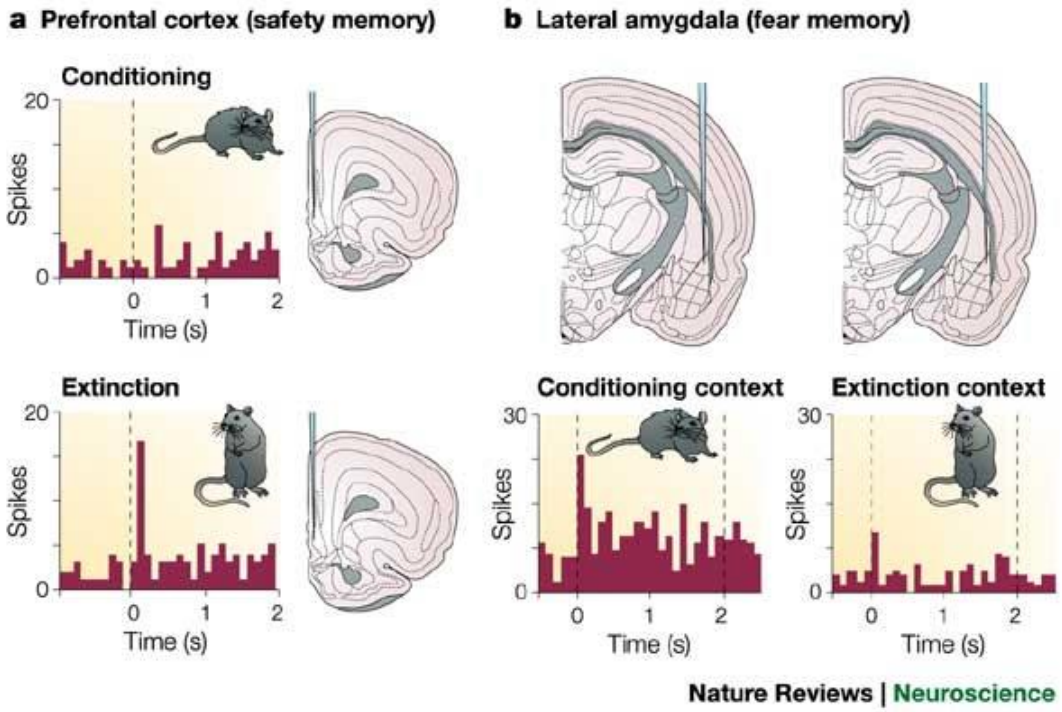
Extinction: a new learning

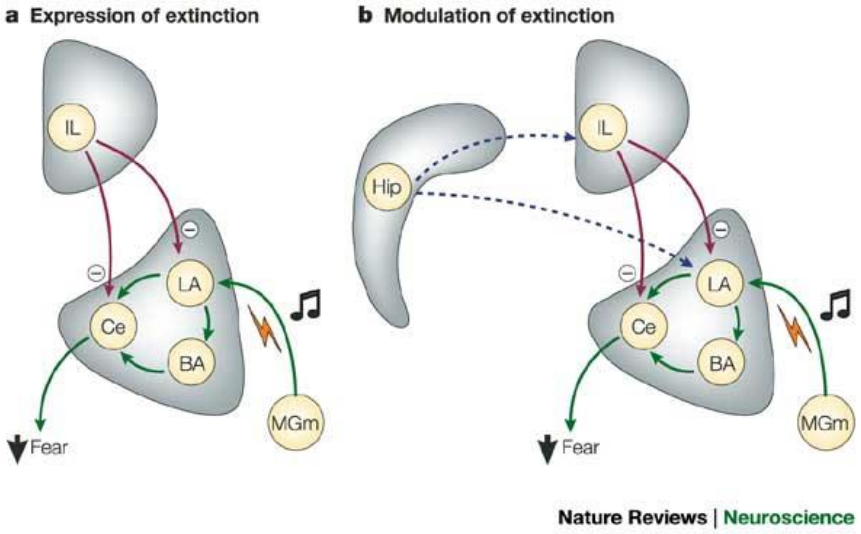


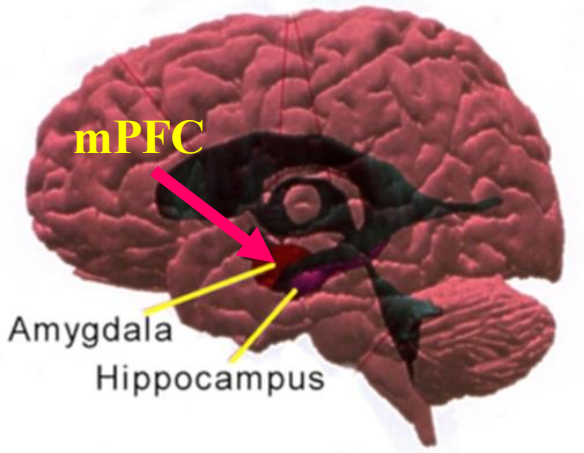


Faster re-learning

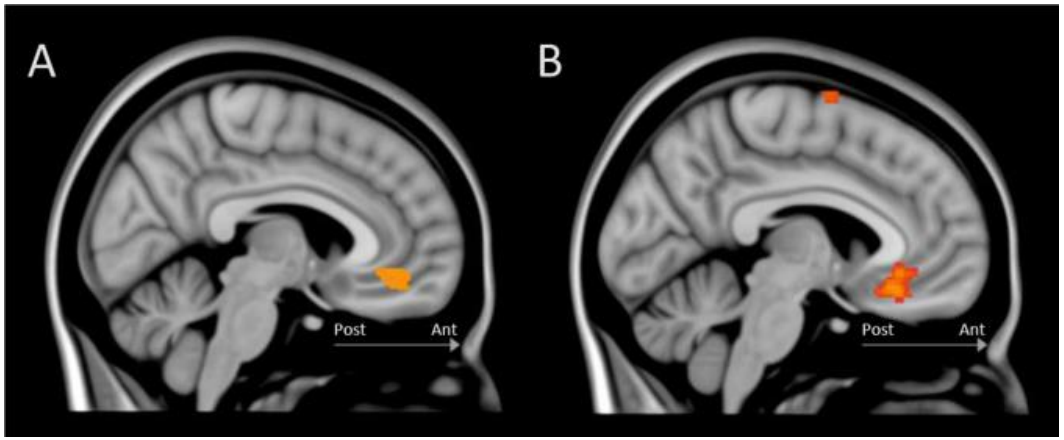
Extinction: brain mechanisms

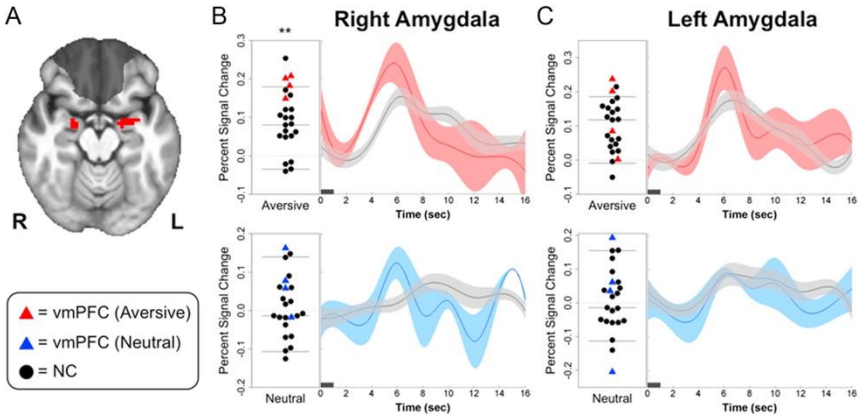


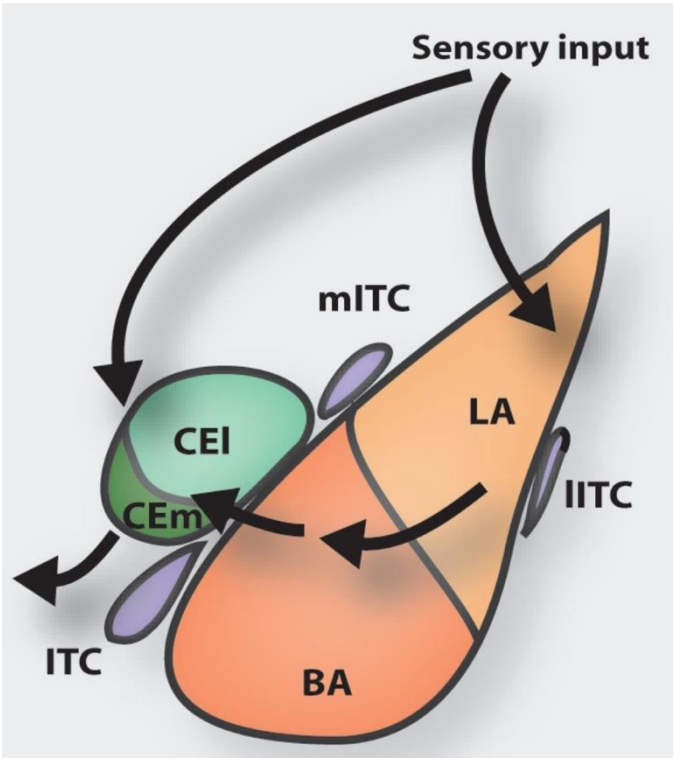




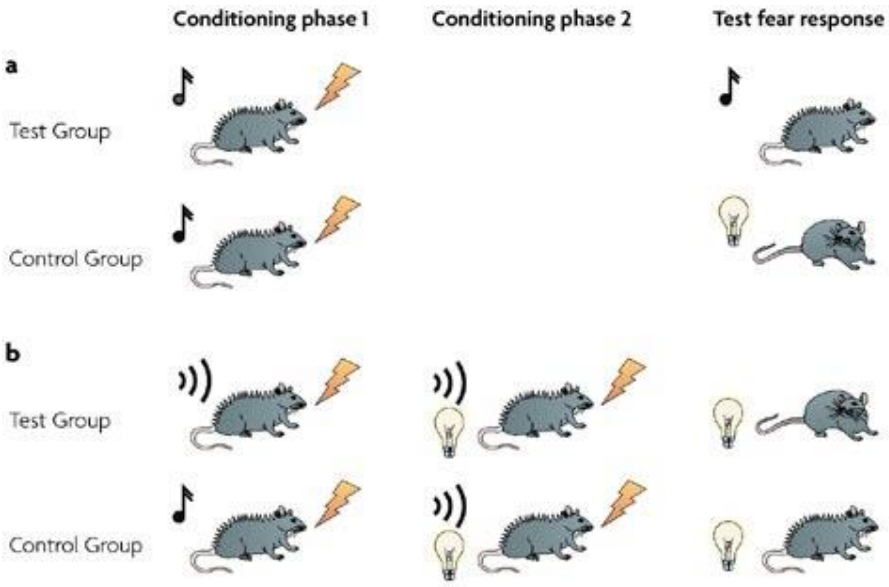
PFC -> Amygdala in extinction







More than contingency: Surprise / added information



Aversive conditioning

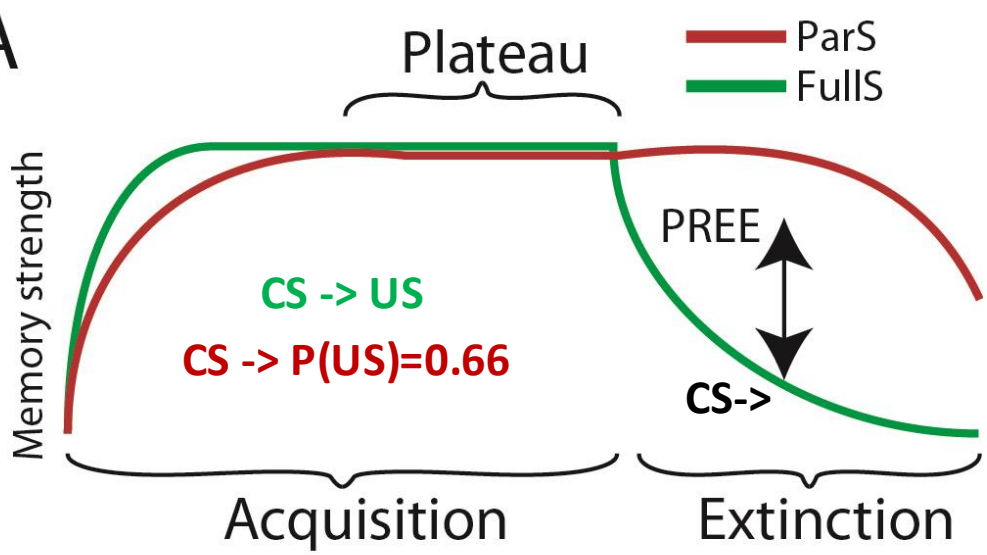
Blocking



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Partial reinforcement extinction effect

- Partial reinforcement
  - Fixed/variable ratio
  - Fixed/variable schedule
- Results in longer extinction

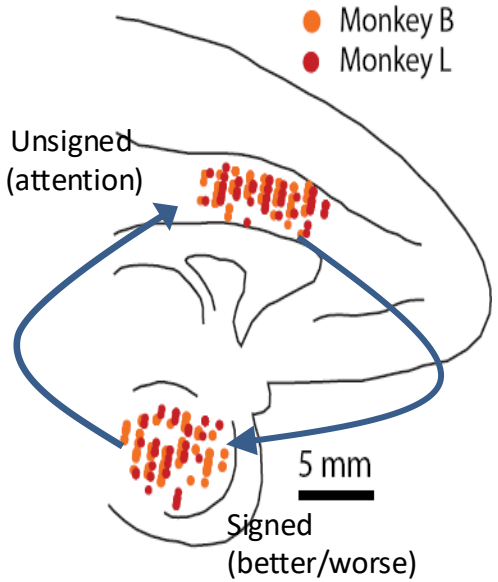


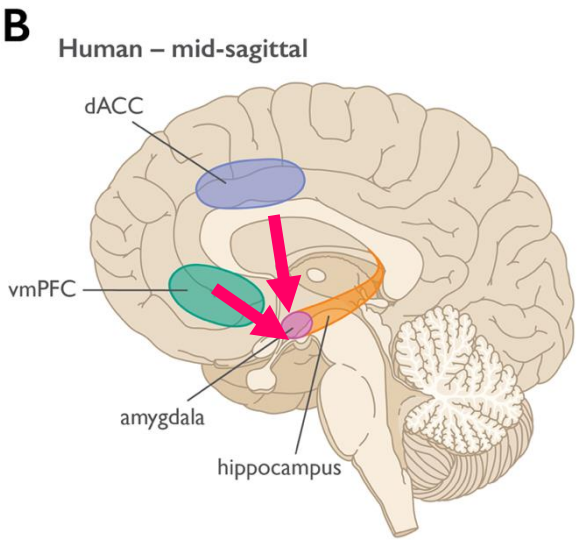
  - Frustration theory (Amsel): The omission of the US induces frustration. Therefore, during extinction, the frustration predicts the US.
  - Sequential theory (Capaldi): conditioning to strings of NNNRNNNR

Teaching signals (reinforcement learning theory): a requirement for safety learning

- *Prediction errors*: the difference between expected and observed

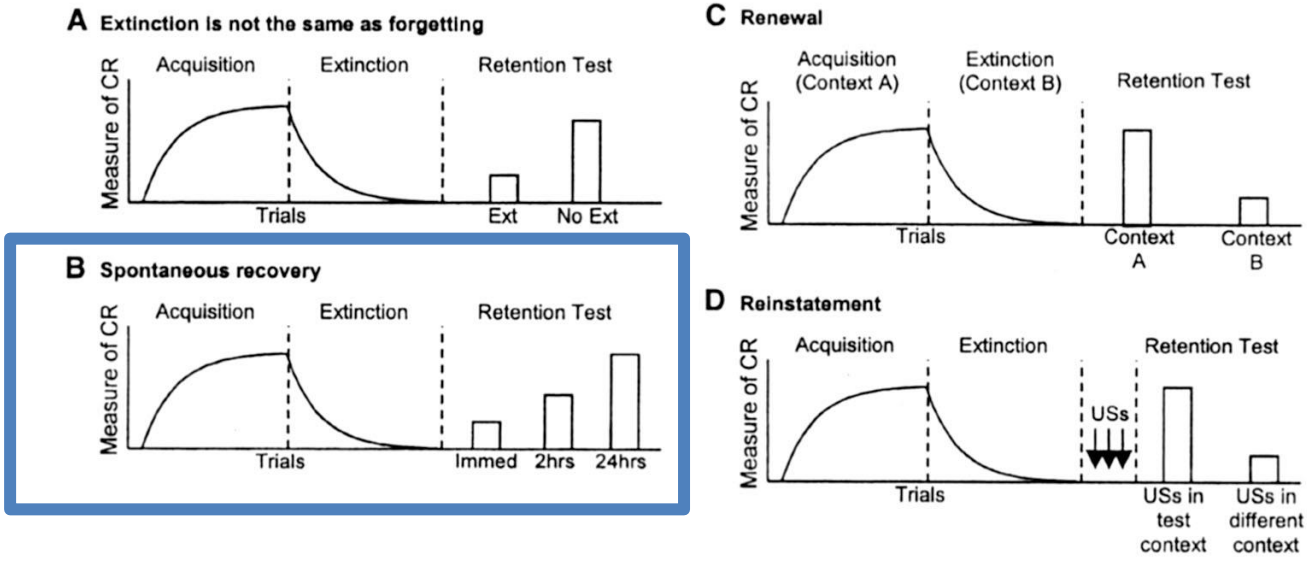
- First: Unsigned/absolute (attention) error signals in the amygdala
- Then: Signed errors in the ACC





Klavir et al, Neuron, 2013

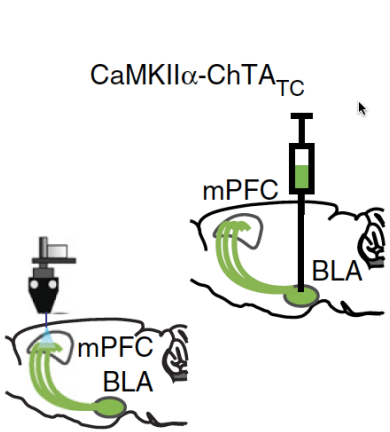
Spontaneous recovery

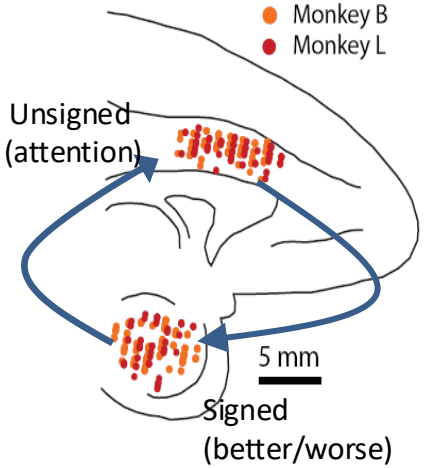


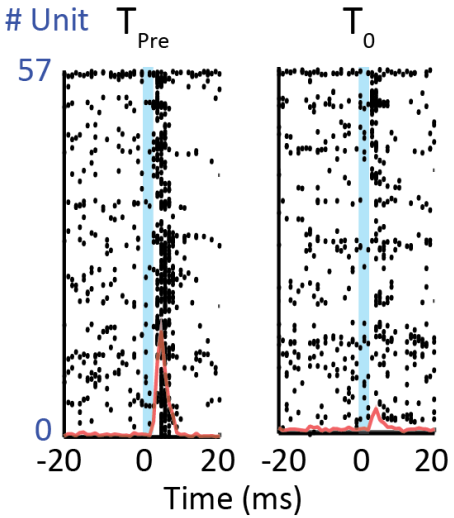
Inducing LTD specifically in amygdala->PFC

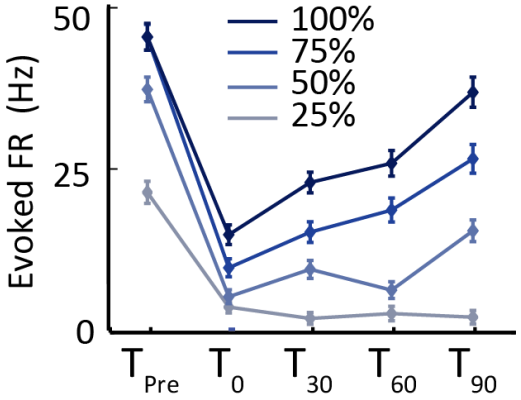
If we attenuate amygdala output to PFC,

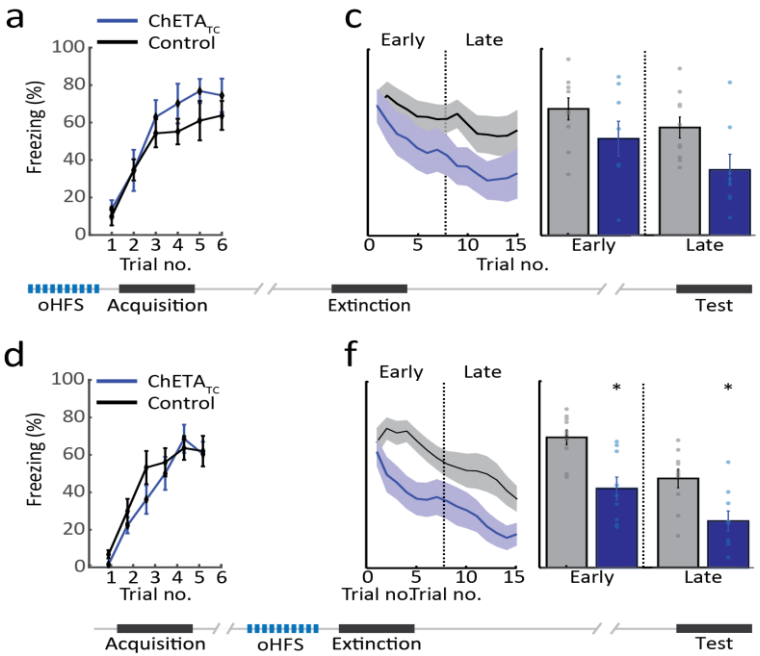
can we facilitate extinction and reduce later memory?







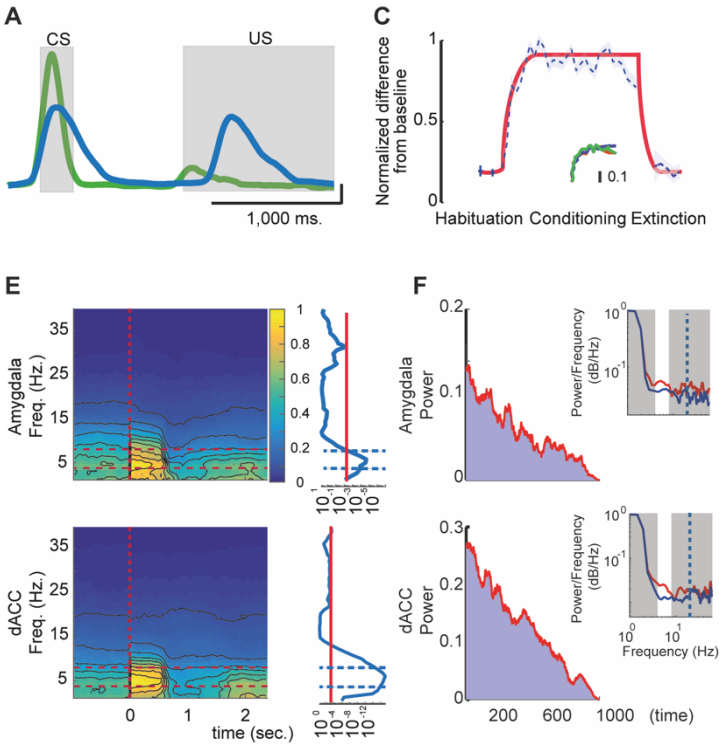


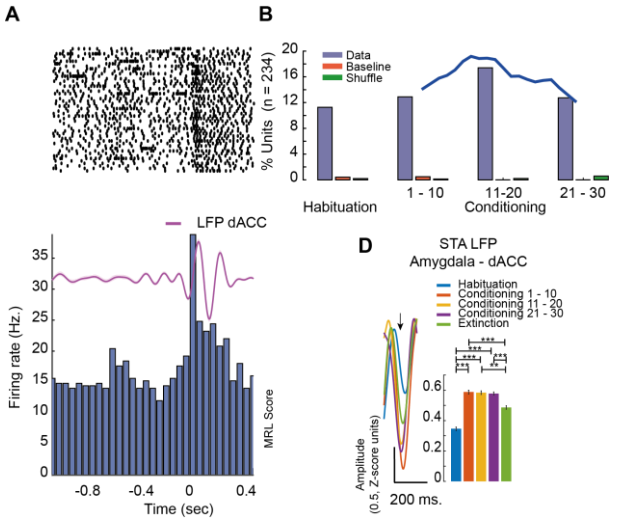


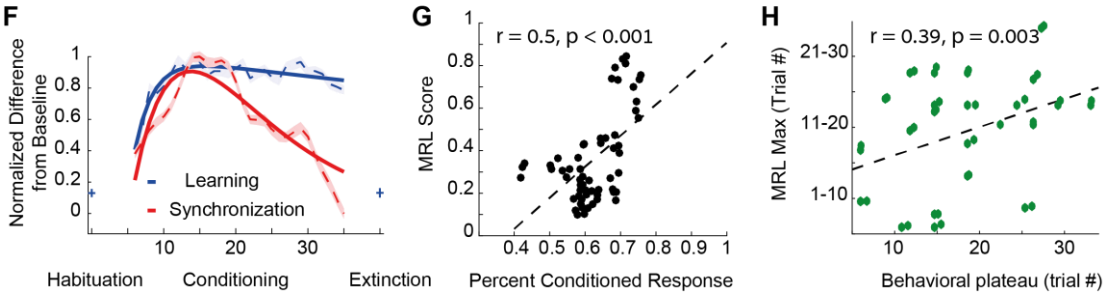
Klavir et al, Nat.Neurosci., 2017

Mechanism of Amygdala->ACC synchronization in the primate

- Theta band (4-8Hz)



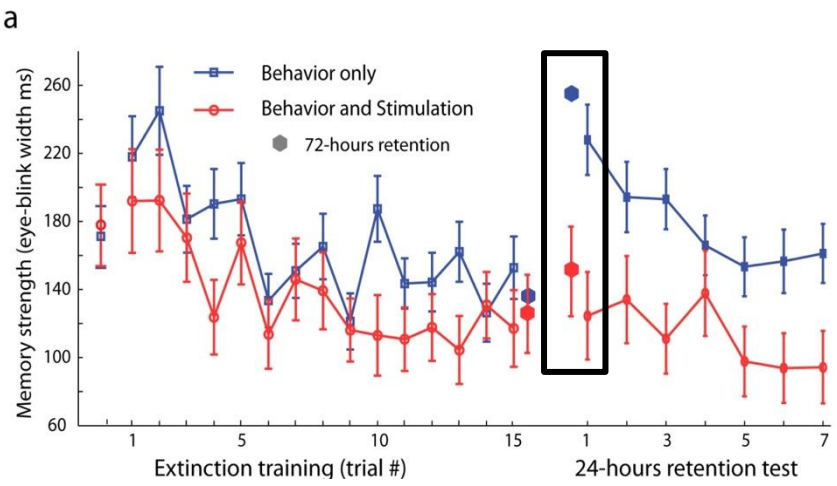




Taub et al, Neuron, 2018

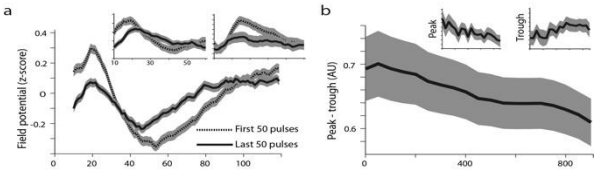
Reducing spontaneous recovery in the primate

- Low Frequency-stimulation (LFS) reduces excitability in the dACC and prevents spontaneous recovery

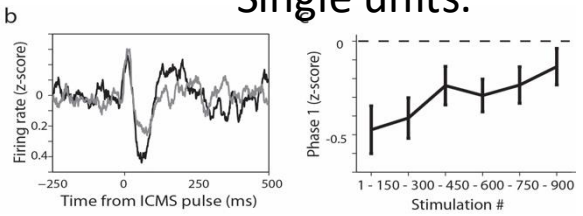


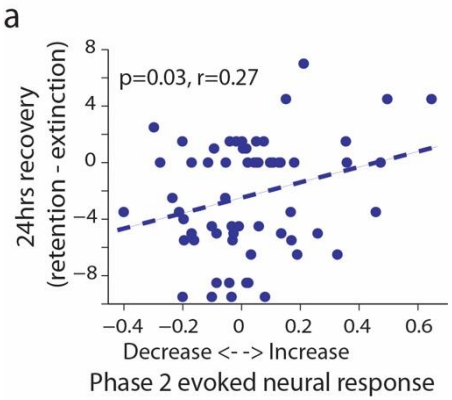
Klavir et al, Nat.Neurosci., 2017

Local fields:



Single units:



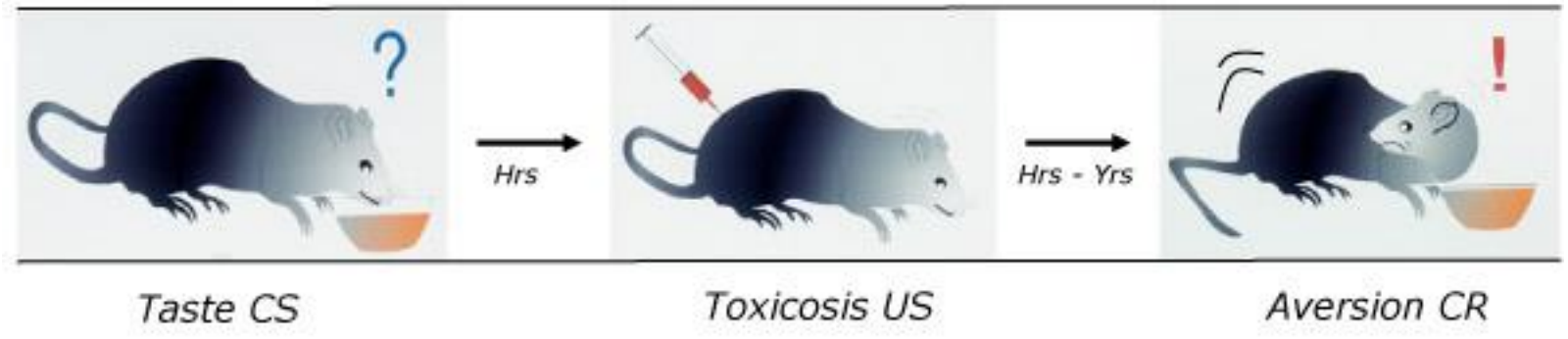


Klavir et al, J. Neurosci., 2012

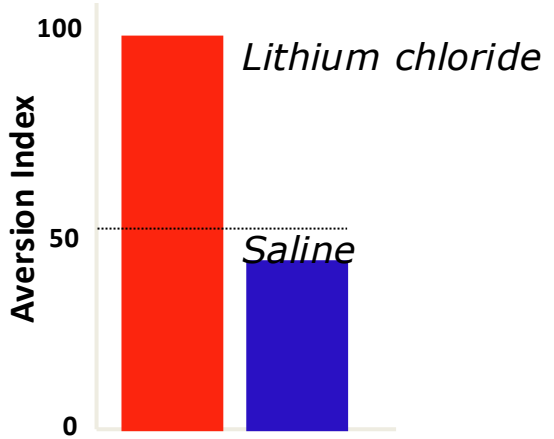
Genud-Gabai et al, J.Neurosci., 2013

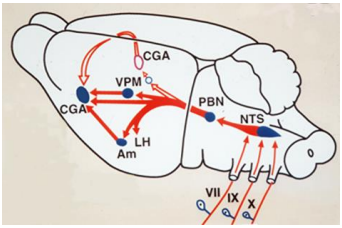
Garcia J

Conditioned Taste Aversion



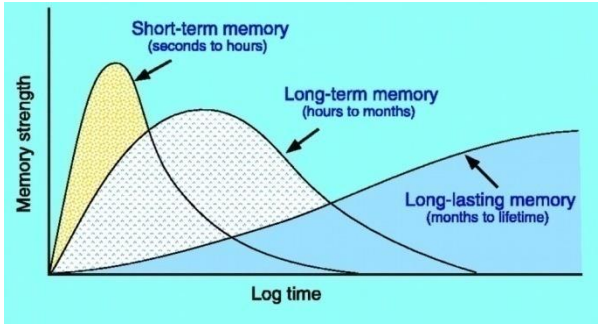
- One-trial learning
- Long-delay learning (few hours)
  - A [lack of] interference effect?
  - Still a problem for neuroscientists

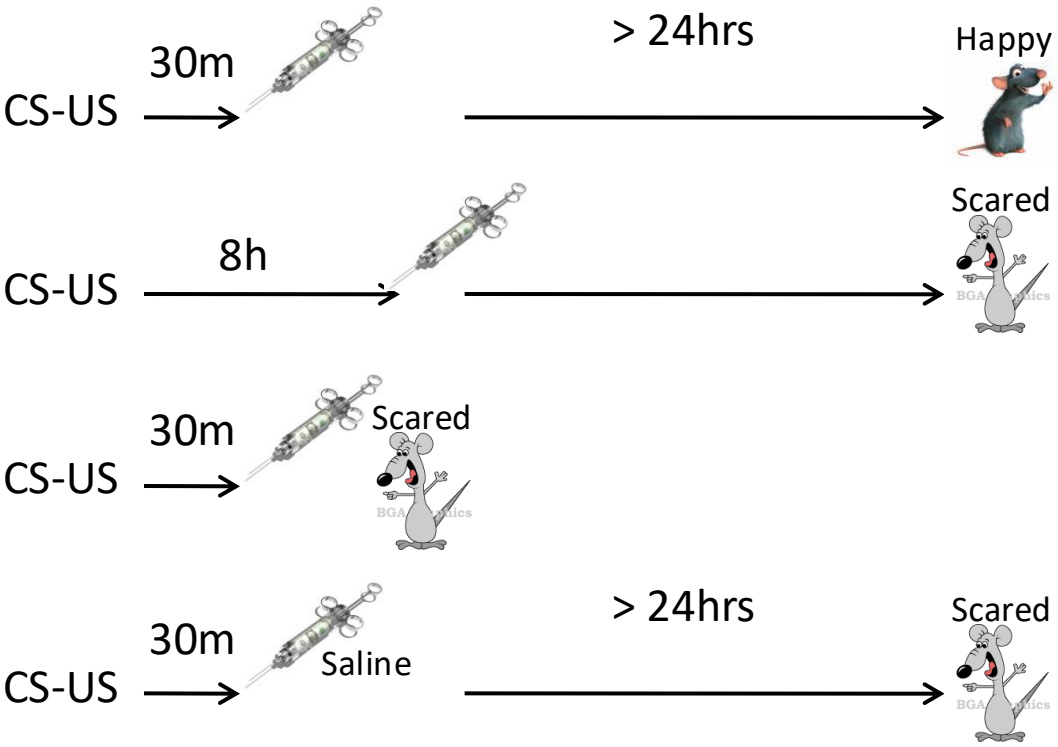


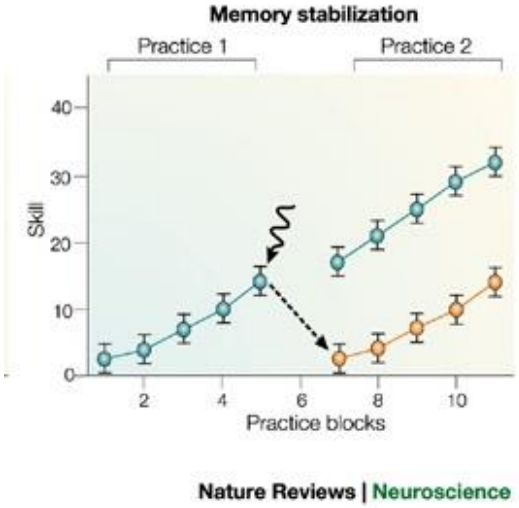


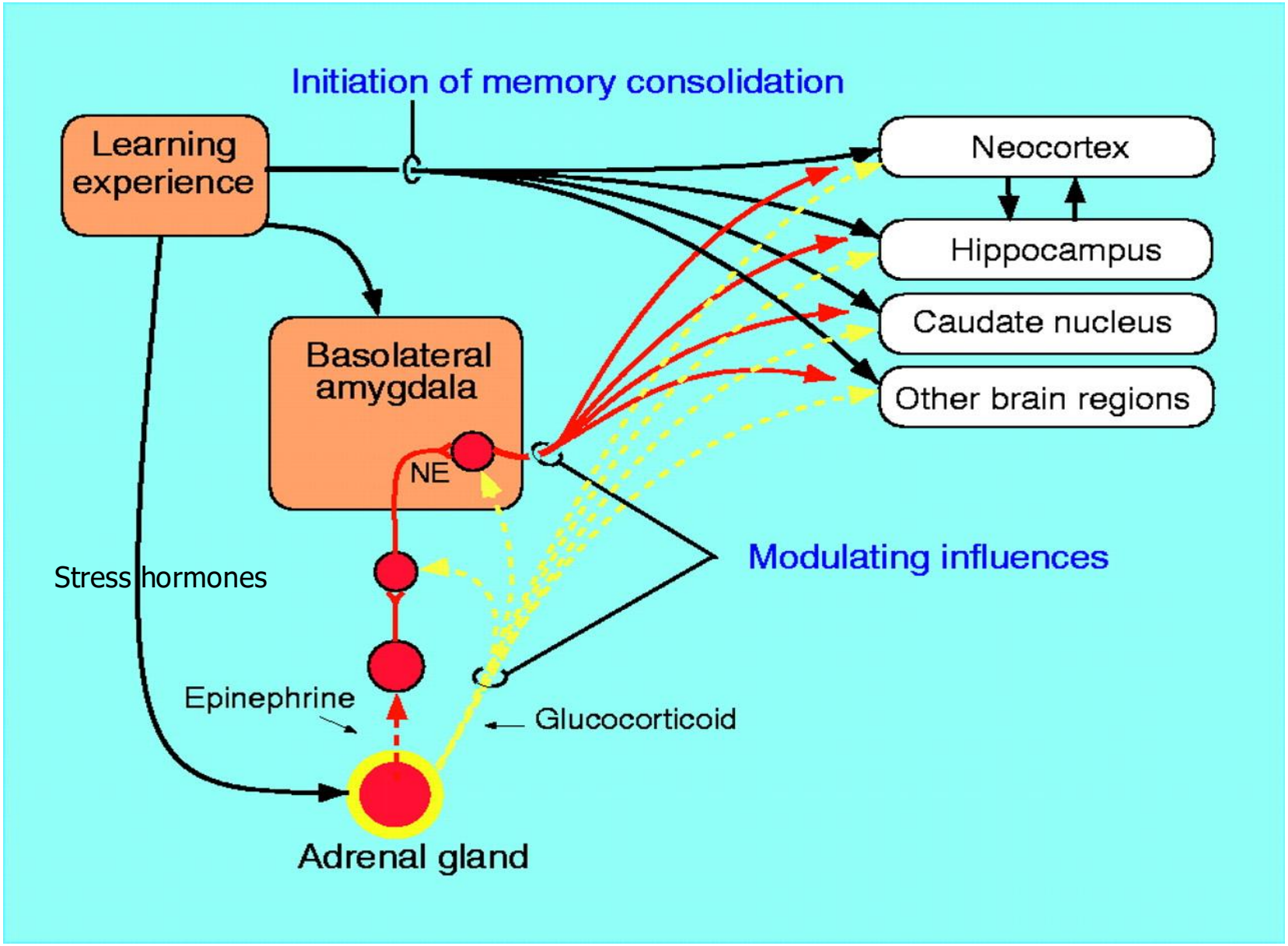
Consolidation

- Anisomycin, a protein synthesis inhibitor, into the Basolateral complex of the amygdala (BLA)
  - No effect on short-term-memory
  - No effect after XX time (rule of thumb is 6hrs)
  - But harms long-term memory below that.





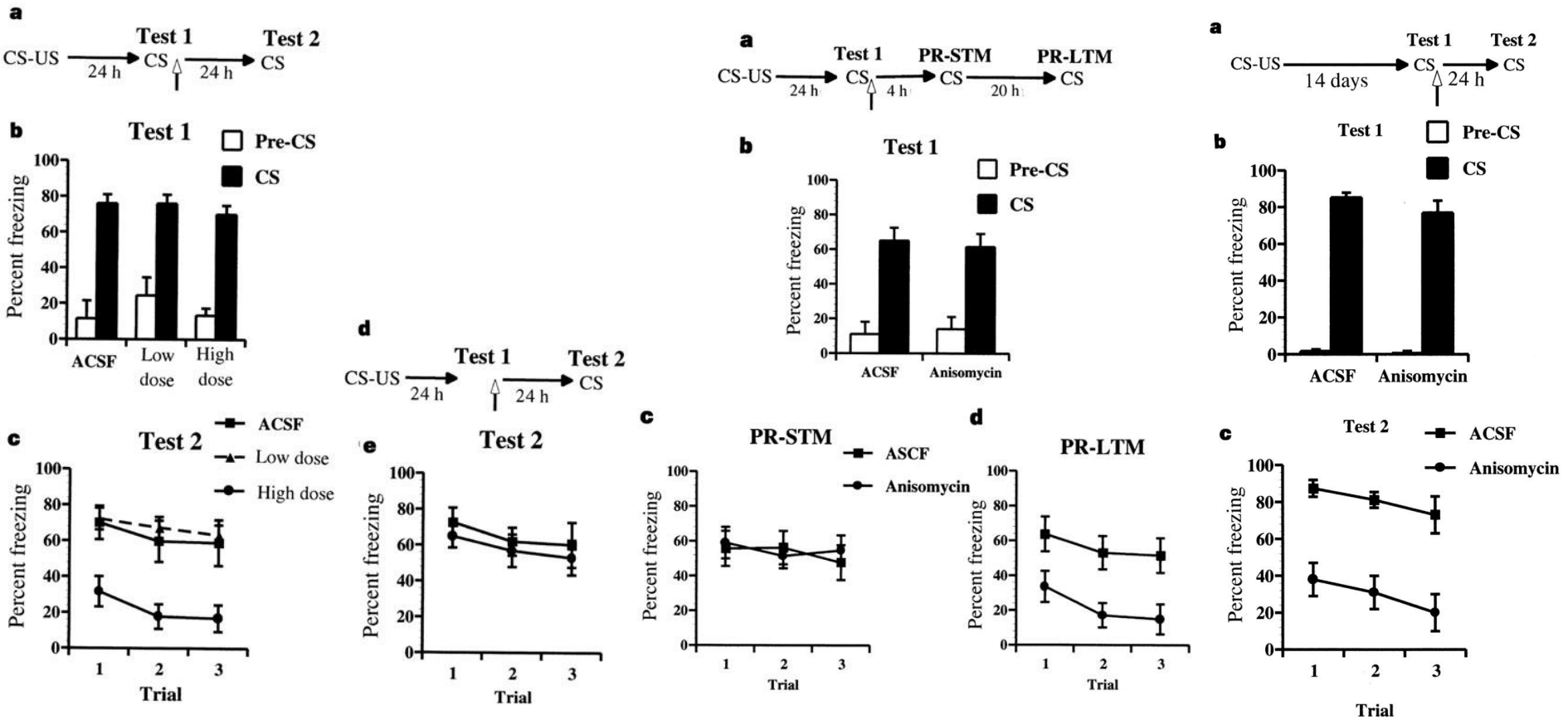




Mcgaugh JL, science, 2000

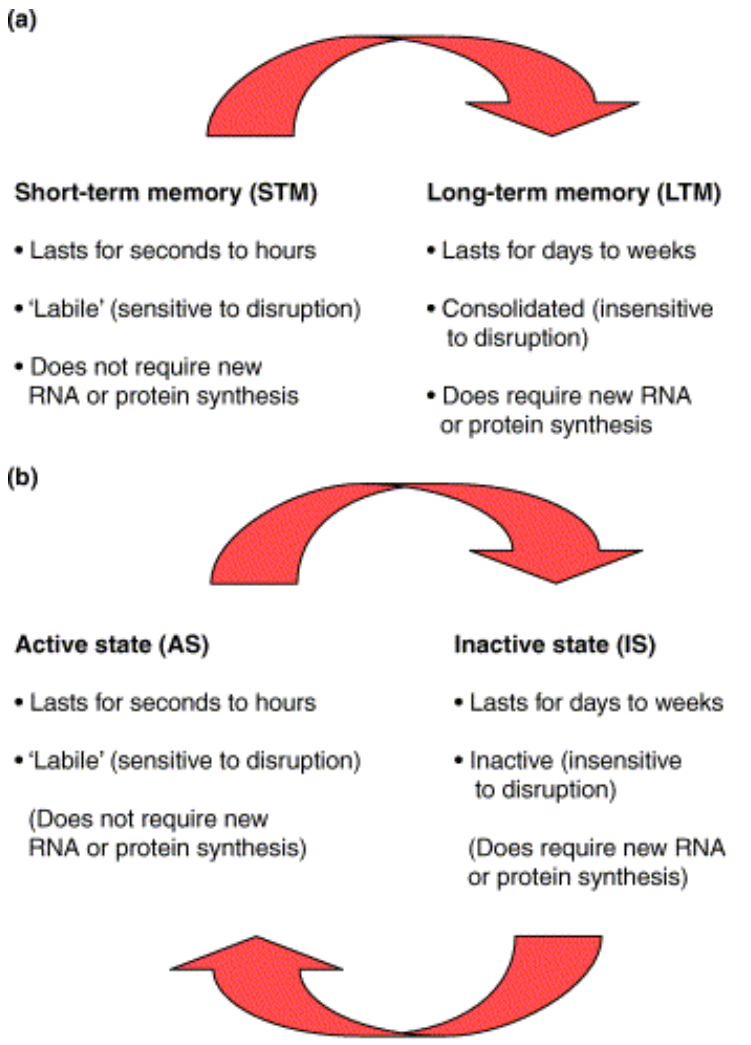
Reconsolidation

No effect on STM



Nader, Ledoux, Nature 2000

An updated view of memories



Memories that over-generalize
(better safe than sorry)



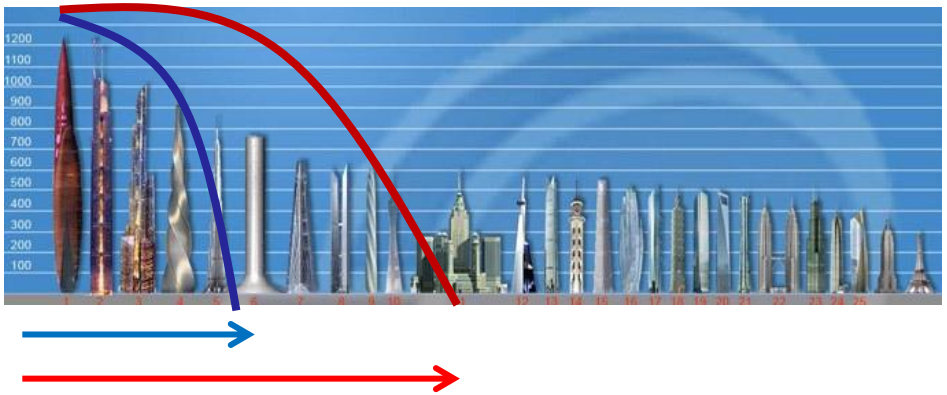
- Little Albert experiment: Watson and Rayner, 1920; One-year old Albert feared a rat that was previously paired with banging of steel rail.
- They then showed he was afraid of other white fury objects as well.
  - Rabbit
  - Watson dressed as Santa-Claus

Was little Albert wrong or right?

Better safe than sorry

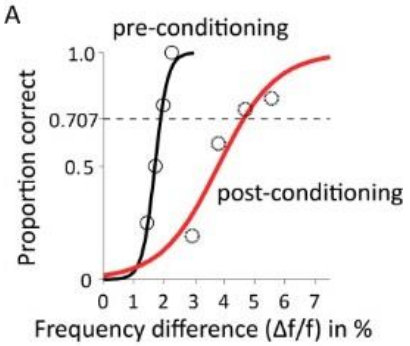
Over-generalization in anxiety/trauma

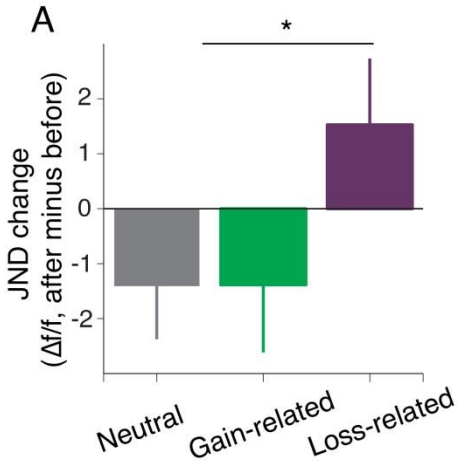




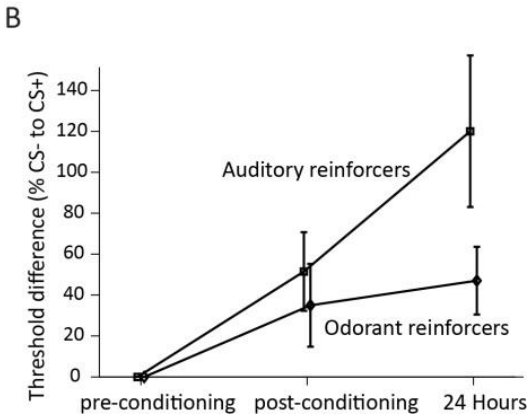
Aversive learning induces perceptual learning: an increase in discrimination thresholds

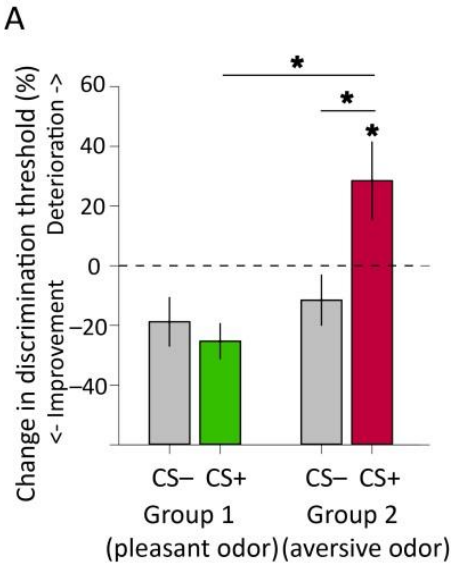
- Increased the Just-noticeable-difference (JND) for the conditioned stimulus
- Persists over-night (24 hours)
- **Robust to different reinforcer modalities (odors, sounds, pictures)**
- **Robust to CS modalities (auditory, visual)**
  basic stimulus features: tone frequency, bar orientation, acuity
- Monetary loss





Resnik et al, Nat. Neurosci. 2011

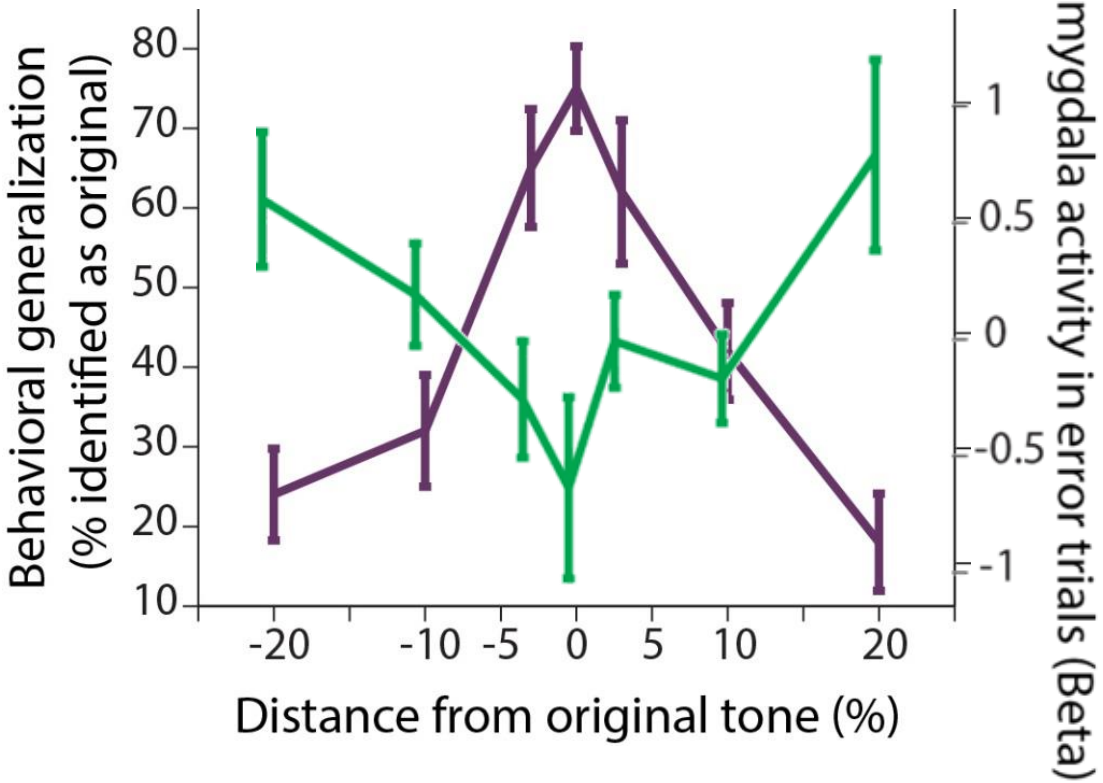


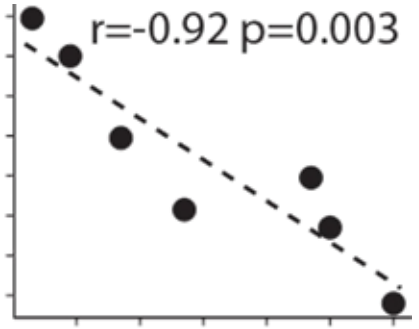


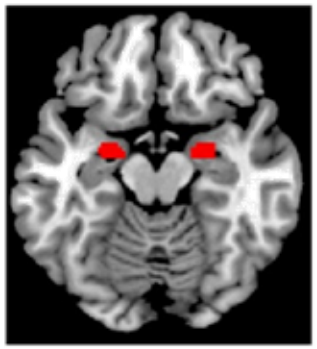
Laufer & Paz, J.Neurosci. 2012

Shalev, Paz, Avidan, J.Neurosci. 2018

Activity in the amygdala is correlated with making a bigger error ("overrides" large perceptual differences)



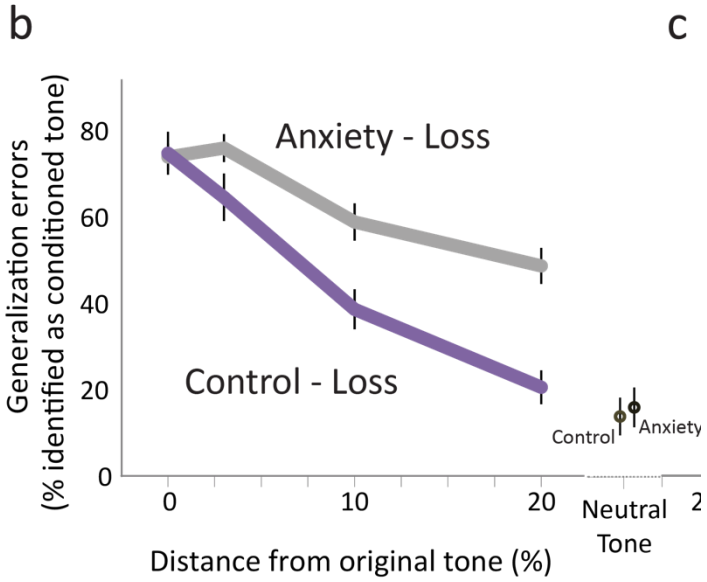




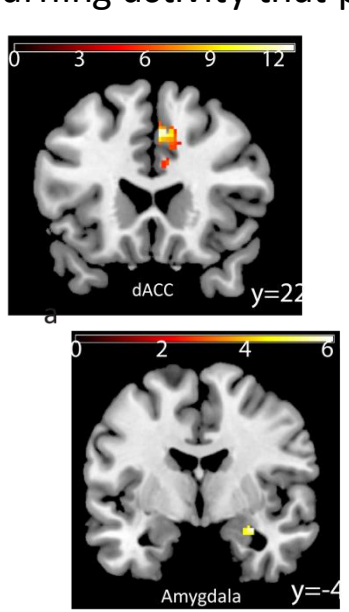
Laufer & Paz, J.Neurosci. 2012

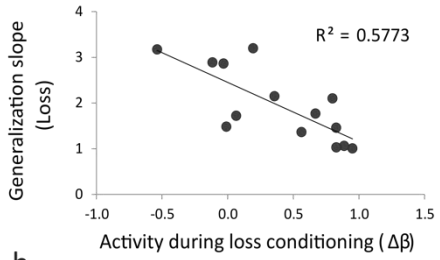
Schechtman et al, J.Neurosci. 2010

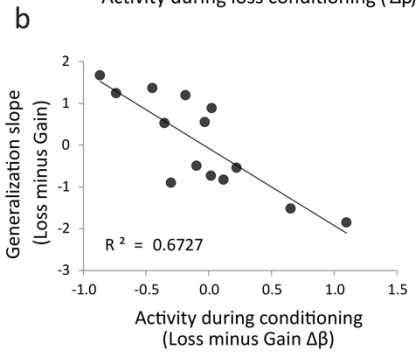
Wider generalization in Anxiety patients



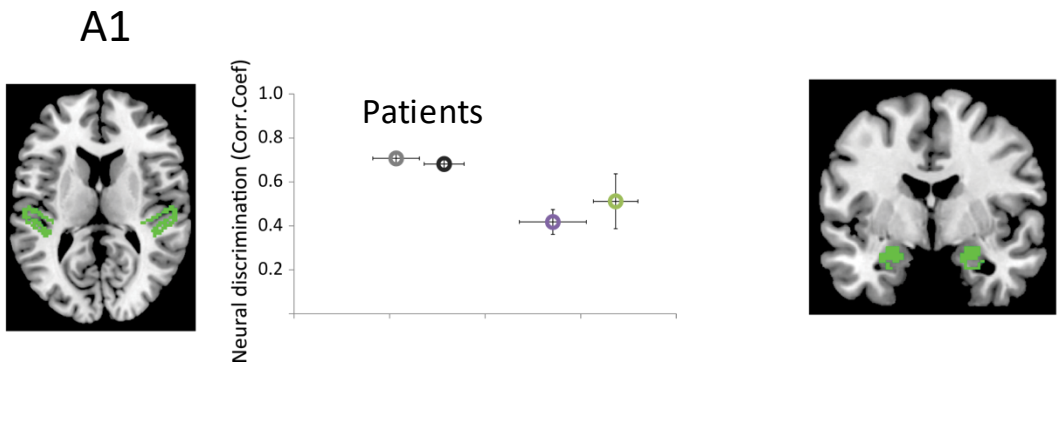
Learning activity that predict later generalization width





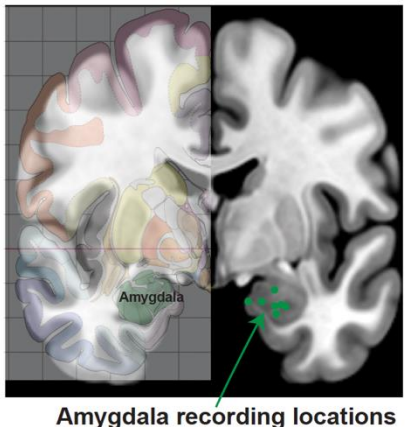


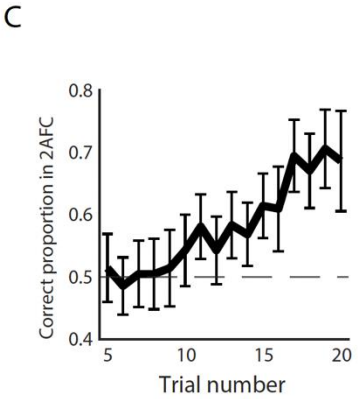
Activation during generalization that matches its width

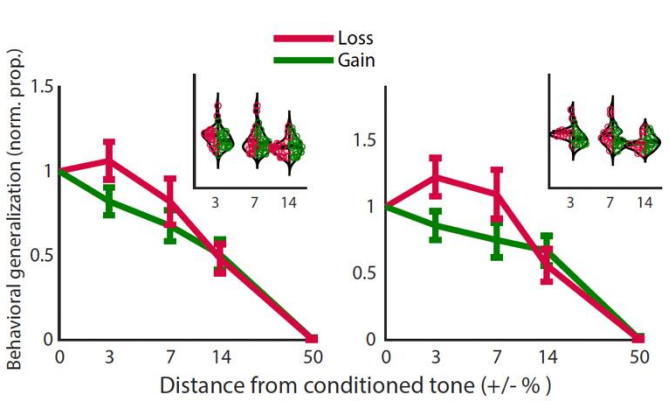


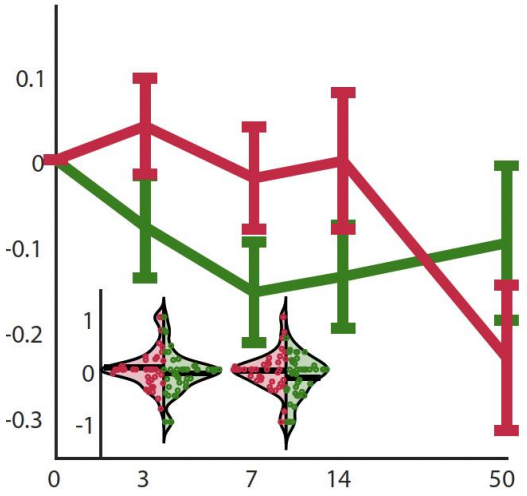
Laufer et al, Curr.Biol.2016

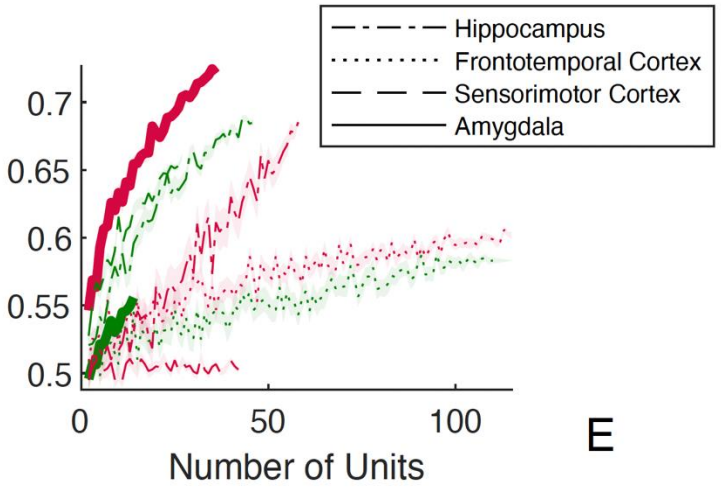
Human amygdala neurons











Take home: extreme (yet rational) aversive learning can lead to PTSD symptoms

- Resistance to extinction
- spontaneous recovery
- Over-generalization
- Exploration-Exploitation tradeoff
- **A delicate balance in information transfer between amygdala-ACC-mPFC (Hippocampus?)**

